Microscopic Foundations of the Curvature Relaxation Model: From Quantum Geometry to Macroscopic Saturation

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(Dated: June 23, 2026)

Papers I and II established the Curvature Relaxation Model (CRM) as a phenomenologically successful alternative to ΛCDM, eliminating the dark sector through geometric curvature return ($\Delta\chi^2 = -5.5$ vs. ΛCDM, joint SN+CMB+BAO, 6 parameters, zero EDE; note: χ^2 computed with diagonal covariance approximation, neglecting off-diagonal correlations). This paper addresses the microscopic foundations: *What quantum system yields the saturation ODE* $d\Omega_\Phi/da = k[1 - (\Omega_\Phi/\Phi_0)^2]$ *and the running coupling* $\beta_{\rm eff}(a)$*?* We derive the effective Lagrangian $\mathcal{L}_{\rm CRM} = R/(16\pi G) + \gamma R^2 + \frac{1}{2}(\partial\phi)^2 - V_{\rm PT}(\phi)$, where the R^2 term generates the geometric "dark matter" contribution and a Pöschl-Teller scalar provides saturation dynamics. The trace coupling $\mathcal{F}(T/\rho)$ ensures BBN protection as a rigorous consequence of conformal symmetry. We explore four UV-completion candidates (scalar double-well, Loop Quantum Gravity, Finsler geometry, information-theoretic spacetime), all producing saturation dynamics of the required form. The running β is interpreted as a second order parameter in a geometric phase transition. Ghost freedom, tachyonic stability, and solar system compatibility (chameleon screening, $\lambda_C^{\rm solar} \sim 20\,\rm m$) are verified. Testable predictions – scalaron Compton wavelength, gravitational wave speed $c_T = c$, lensing parameter, and CMB power spectrum modifications – are independent of the UV completion. The ontological picture reduces to spacetime curvature in three phases plus baryonic matter.

I. INTRODUCTION: THE CENTRAL QUESTION

The Curvature Relaxation Model (CRM) [1] and its MOND-compatible extension [2] have demonstrated remarkable phenomenological success:

- **Paper I:** The standard CRM replaces dark energy with a curvature return potential, achieving $\Delta\chi^2 = -12.2$ vs. ΛCDM on Pantheon+ data.
- **Paper II:** The extended CRM replaces the particle dark sector with spacetime geometry, achieving a universe with exclusively baryonic matter content. The SN-only analysis yields $\Delta\chi^2 = -26.3$ with $\beta = 2.02 \pm 0.20$ (curvature scaling). Crucially, a *running curvature coupling* $\beta_{\rm eff}(a)$ – transitioning from CDM-like ($\beta_{\rm early} \approx 2.82$) at $z > 6$ to curvature-like ($\beta \approx 2.0$) at low z – combined with a scale-dependent MOND coupling $\mu(a)$ achieves joint SN + CMB + BAO compatibility: $\ell_A = 301.471$ (Planck: 301.471), $\mathcal{R} = 1.7502$ (Planck: 1.7502), $H_0 = 67.3\,\rm km/s/Mpc$, and $\Delta\chi^2 = -5.5$ vs. ΛCDM (preferred zero-EDE variant).

Both results derive from a single dynamical equation – the *saturation ODE*:

$$\frac{d\Omega_\Phi}{da} = k\left[1 - \left(\frac{\Omega_\Phi}{\Phi_0}\right)^2\right] \tag{1}$$

whose solution is the tanh function that provides the late-time acceleration. The extended model adds a power-law term $\alpha \cdot a^{-\beta_{\rm eff}}$ representing the unsaturated (early-time) phase of the same geometric process, with a running coupling $\beta_{\rm eff}(a)$ that encodes the curvature-dependent transition between gravitational regimes.

The central question of this paper is:

> *Which microscopic (quantum) system has the property that its macroscopic (thermodynamic) limit yields (i) the saturation ODE* (1)*, (ii) the running coupling* $\beta_{\rm eff}(a)$*, and (iii) the full extended Friedmann equation? Can the entire framework be derived from a Lagrangian?*

This question is not merely academic. Without a Lagrangian formulation, the CRM cannot:

1. Be consistently coupled to matter fields
2. Generate perturbation equations for C_ℓ and $P(k)$ predictions
3. Be connected to known quantum gravity frameworks
4. Be considered a complete physical theory

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† AI tools (Claude, Anthropic; Copilot, Microsoft/GitHub; Gemini, Google DeepMind) were used for mathematical formalization, code development, and text generation. All physical hypotheses, scientific interpretation, and responsibility for the content lie solely with the author. The analysis code is available in the GitHub repository.; Paper III in the CRM series.

II. THE EFFECTIVE LAGRANGIAN

A. Requirements

The effective Lagrangian $\mathcal{L}_{\mathrm{CRM}}$ must satisfy:

1. **Background:** The Euler-Lagrange equations, evaluated on the FLRW metric, must yield the extended Friedmann equation:
$$H^2(a) = H_0^2\left[\Omega_b\, a^{-3} + \Phi_0 \cdot f_{\mathrm{sat}}(a) + \alpha \cdot a^{-\beta}\right] \tag{2}$$
2. **Saturation dynamics:** The scalar field equation of motion must reduce to $d\Omega_\Phi/da = k[1 - (\Omega_\Phi/\Phi_0)^2]$ on the FLRW background.
3. **General covariance:** The action must be diffeomorphism-invariant.
4. **Correct limits:** In the limit $k \to 0$, $\alpha \to 0$, the theory must reduce to GR with cosmological constant.

B. Scalar Field Approach

The most natural Lagrangian formulation introduces a scalar field ϕ with a potential $V(\phi)$:

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi - V(\phi) + \mathcal{L}_m\right] \tag{3}$$

For the saturation ODE to emerge, we require $V(\phi)$ such that the homogeneous field equation on FLRW yields tanh-type solutions.

Proposition 1 (Double-Well Saturation Potential) *The potential*

$$V(\phi) = V_0\left[1 - \tanh^2\left(\frac{\phi}{\phi_0}\right)\right] = \frac{V_0}{\cosh^2(\phi/\phi_0)} \tag{4}$$

produces a scalar field equation whose late-time solution on the FLRW background is $\phi(a) \propto \tanh(k(a - a_{\mathrm{trans}}))$, reproducing the saturation term of the CRM.

Sketch of proof: The Klein-Gordon equation on FLRW,

$$\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0 \tag{5}$$

with $V'(\phi) = -2V_0 \tanh(\phi/\phi_0)/(\phi_0\cosh^2(\phi/\phi_0))$, admits the solution $\phi = \phi_0\tanh(\lambda t)$ in the slow-roll regime where $\ddot{\phi} \ll 3H\dot{\phi}$, with λ related to k and H_0. The energy density $\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi)$ then maps to $\Omega_\Phi(a) = \Phi_0 \cdot f_{\mathrm{sat}}(a)$. □

Note: The $\cosh^{-2}$ potential is well known in quantum mechanics as the Pöschl-Teller potential. Its appearance here suggests a connection between quantum bound states and cosmological saturation.[1]

Philosophical tension: ϕ as an additional scalar field. The introduction of an additional scalar field ϕ appears to contradict the CRM's central claim of eliminating the dark sector. The standard dark energy critique applies: a quintessence-type field with a tuned potential is structurally similar to a dark energy component, even if it is not called one. Three responses can be offered:

(i) Qualitative difference: ϕ is not a freely adjustable phenomenological fluid. It has a concrete Lagrangian potential (Eq. 4), a known quantum-mechanical analog (Pöschl-Teller), and its dynamics are fixed by two parameters (V_0, ϕ_0). Dark energy models typically require an equation of state $w(a)$ parametrized by additional free functions.

(ii) Derivation from geometry (preferred route): The $f(R) = R + 2\gamma R^2$ formulation (Section II D) avoids introducing ϕ explicitly: by conformal equivalence, the scalaron *is* ϕ, emerging as a geometric degree of freedom from the R^2 term, not as an independently postulated field. The scalar field approach (this subsection) and the $f(R)$ approach are mathematically equivalent.

(iii) Acknowledged limitation: The reviewer's objection is valid at the following level: adding any new degree of freedom to GR—whether called "scalaron," "scalar field," or "geometric term"—does expand the field content beyond baryons + photons + neutrinos. The claim that "no dark sector exists" must therefore be read as: "no *particle physics* dark sector (no dark matter particles, no cosmological constant)"—not as "no additional fields beyond the Standard Model metric." The CRM replaces particle-physics unknowns with geometric unknowns. Whether this is considered an elimination or a transformation of the dark sector is a matter of framing; we believe the geometric origin provides greater theoretical predictability, but acknowledge the objection's force.

C. The Power-Law Term: Geometric Origin

The geometric "dark matter" term $\alpha \cdot a^{-\beta}$ with $\beta \approx 2$ requires a separate origin. Two approaches are possible:

Approach 1: Curvature-squared terms. Adding a Gauss-Bonnet or R^2 term to the action:

$$S_{\mathrm{geom}} = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} + \gamma\, R^2 + \delta\, R_{\mu\nu}R^{\mu\nu}\right] \tag{6}$$

[1] The Pöschl-Teller potential has recently found renewed application in black hole physics, where it provides analytically tractable quasinormal mode spectra for Schwarzschild and Nariai spacetimes [57, 58]. That the same potential governs both cosmological saturation (CRM) and gravitational ringdown underscores a deeper connection between bound-state quantum mechanics and spacetime dynamics.

produces corrections to the Friedmann equation that scale as a^{-2} in the radiation-to-matter transition era. The coefficient γ can be related to α.

Approach 2: Vector field (AeST-inspired). Following Skordis & Złośnik [6], a timelike vector field A_μ constrained by $g^{\mu\nu}A_\mu A_\nu = -1$ contributes an effective energy density that scales non-standardly with a. The CRM power-law term may emerge as the cosmological background of such a vector field.

D. The Combined Action

Combining both contributions, the full CRM action reads:

$$S_{\mathrm{CRM}} = \int d^4x\sqrt{-g}\Big[\frac{R}{16\pi G} + \gamma R^2 - \frac{1}{2}(\partial\phi)^2 - \frac{V_0}{\cosh^2(\phi/\phi_0)} + \mathcal{L}_m\Big] \tag{7}$$

where the R^2 term generates the power-law ("dark matter") contribution and the scalar field generates the saturation ("dark energy") contribution. The equilibrium between null space and spacetime bubble (Paper I) is encoded in the balance between γ and V_0.

A crucial refinement introduced in Paper II [2] is the *trace coupling*. We now demonstrate that this is *not* an ad hoc postulate but a direct consequence of the R^2 action. Taking the trace of the field equations for $f(R) = R + 2\gamma R^2$ gives:

$$R + 12\gamma\,\Box R = -8\pi G\,T \tag{8}$$

where $T = g^{\mu\nu}T_{\mu\nu}$ is the energy-momentum trace. During the radiation era, conformal symmetry demands $T_{\mathrm{rad}} = -\rho_r + 3p_r = 0$ (since $w = 1/3$). Equation (8) then reduces to $R + 12\gamma\,\Box R = 0$, whose FLRW solution is a decaying mode: $R \to 0$ as $a \to 0$. Since the scalaron (geometric DM) is sourced by R^2, it vanishes identically when $R = 0$. The geometric DM contribution is therefore *automatically* suppressed during the radiation era – BBN is protected without any additional mechanism.

The only genuine *postulate* is the choice of Lagrangian $f(R) = R + 2\gamma R^2$; the suppression factor $\mathcal{S}(a) = 1/(1 + a_{\mathrm{eq}}/a)$ of Paper II is a phenomenological parametrization of this rigorous result. The modified action with the full coupling reads:

$$S_{\mathrm{CRM}} = \int d^4x\sqrt{-g}\Big[\frac{R}{16\pi G} + \gamma\,\mathcal{F}(T/\rho)\,R^2 - \frac{1}{2}(\partial\phi)^2 - \frac{V_0}{\cosh^2(\phi/\phi_0)} + \mathcal{L}_m\Big] \tag{9}$$

where $\mathcal{F}(T/\rho) = |T|/(|T| + \rho_{\mathrm{rad}}) \to 0$ in the radiation era and $\mathcal{F} \to 1$ in the matter era.

E. Ghost Freedom and Stability

A critical consistency requirement for any higher-derivative theory is the absence of Ostrogradsky ghosts [49]. We now verify that the action (7) satisfies all stability conditions.

Conformal equivalence. The gravitational sector $f(R) = R+2\gamma R^2$ (with $\epsilon = 16\pi G\gamma$) is conformally equivalent to Einstein gravity plus a massive scalar field (the *scalaron*) χ:

$$S = \int d^4x\sqrt{-g_E}\left[\frac{R_E}{16\pi G} - \frac{1}{2}(\partial\chi)^2 - U(\chi)\right] \tag{10}$$

where $\chi = \sqrt{3/(16\pi G)}\ln f_R$ and $U(\chi) = (Rf_R - f)/(2f_R^2)$. This establishes that the theory propagates $2+1+1 = 4$ degrees of freedom (graviton + scalaron + Pöschl-Teller scalar), all with positive kinetic energy.

Proposition 2 (Ghost Freedom of the CRM Action) *The action (7) is ghost-free under the following conditions, all satisfied by construction:*

1. ***No Ostrogradsky ghost:*** $f_{RR} = 2\epsilon > 0$ *(since* $\gamma > 0$*), ensuring positive kinetic energy for the scalaron. QED.*
2. ***No tachyonic instability:*** *The scalaron mass* $m_s^2 = 1/(6\epsilon) > 0$ *for* $\gamma > 0$*. The potential* $U(\chi) \geq 0$ *has a stable minimum at* $\chi = 0$*.*
3. ***No gradient instability:*** *The scalaron sound speed* $c_s^2 = 1$ *in* $f(R)$ *theories (tensor speed* $c_T = c$ *is guaranteed by* $\alpha_T = 0$*).*
4. ***Positive-definite kinetic matrix:*** *The two-field system* (χ, ϕ) *has kinetic matrix* $K = \mathrm{diag}(1,1)$ *in the Einstein frame.*

Trace coupling and stability. The coupling function $\mathcal{F}(T/\rho)$ modifies only the *effective mass* of the scalaron, not its kinetic structure:

$$m_{\mathrm{eff}}^2(a) = \frac{1}{24\gamma\,\mathcal{F}(a)} \tag{11}$$

Since $\mathcal{F} \in [0,1]$ is monotonic and bounded, the mass remains real and positive at all times. At early times ($\mathcal{F} \to 0$), $m_{\mathrm{eff}} \to \infty$ and the scalaron is frozen out. At late times ($\mathcal{F} \to 1$), $m_{\mathrm{eff}} = m_s$.

Newtonian limit and chameleon screening. In the weak-field limit, the scalaron produces a Yukawa correction:

$$V(r) = -\frac{GM}{r}\left(1 + \frac{1}{3}\,e^{-m_{\mathrm{eff}}\,r}\right) \tag{12}$$

Solar system constraints require $m_{\mathrm{eff}}\,r_{\mathrm{AU}} \gg 1$. The trace coupling provides a natural chameleon mechanism: in dense environments ($\rho \gg \rho_{\mathrm{cosm}}$), the effective mass increases as $m_{\mathrm{eff}} \propto \sqrt{\rho_{\mathrm{local}}/\rho_{\mathrm{cosm}}}$. For the Sun ($\rho \sim 1400\,\mathrm{kg/m^3}$), $m_{\mathrm{eff}}^{\mathrm{solar}}/m_s \sim 4 \times 10^{14}$, yielding $\lambda_C^{\mathrm{solar}} \sim 20\,\mathrm{m} \ll 1\,\mathrm{AU}$. The scalaron is thus screened in the solar system for $\gamma \geq \mathcal{O}(1)\,H_0^{-2}$.

1. Scalaron Mass and Local Gravity Tests

From the MCMC posterior (Section VII), $\alpha_{M,0} = 0.0011^{+0.0010}_{-0.0006}$ at the present epoch. The scalaron mass in the cosmological background is determined by the second derivative of the $f(R)$ function:

$$m_s^2 = \frac{R_0}{3\, f_{RR}} = \frac{R_0}{12\gamma}\,, \qquad \gamma \geq \mathcal{O}(1)\, H_0^{-2}\,, \tag{13}$$

where $R_0 \approx 3H_0^2(4\Omega_\Lambda + \Omega_m) \approx 9.2\, H_0^2$ is the background curvature today. Using the constraint $\gamma \sim H_0^{-2}$ (implied by $\alpha_{M,0} \sim 10^{-3}$), this yields:

$$m_s \sim \sqrt{\frac{9.2\, H_0^2}{12\, H_0^{-2}}} \sim 0.88\, H_0^2 \sim \mathcal{O}(10)\, H_0 \sim 10^{-32}\,\mathrm{eV}\,, \tag{14}$$

corresponding to a cosmological Compton wavelength $\lambda_C = 2\pi/m_s \gtrsim \mathcal{O}(100)\,\mathrm{Mpc}$.

Scale-dependent gravitational strength. On sub-Compton scales ($k \gg a\, m_s$), the scalaron mediates a fifth force, enhancing gravity to $\mu \to 4/3$ (the $f(R)$ asymptotic value). On super-Compton scales ($k \ll a\, m_s$), the scalaron is too heavy to propagate, and GR is recovered ($\mu \to 1$). The transition scale $\lambda_C \sim 100\,\mathrm{Mpc}$ is comparable to the baryon acoustic oscillation scale, ensuring compatibility with large-scale structure observations.

Local gravity tests. The chameleon mechanism [40] screens the scalaron in dense environments. In the solar system, the local curvature $R_{\mathrm{solar}} \sim 8\pi G\rho_\odot \gg R_0$ increases the effective scalaron mass according to:

$$\begin{aligned} m_{\mathrm{eff}}^2(\rho) &= \frac{R(\rho)}{12\gamma} \\ \Rightarrow \quad \frac{m_{\mathrm{eff}}^{\mathrm{solar}}}{m_s} &\sim \sqrt{\frac{R_{\mathrm{solar}}}{R_0}} \sim \sqrt{\frac{8\pi G\rho_\odot}{9.2\, H_0^2}} \sim 4 \times 10^{14}\,, \end{aligned} \tag{15}$$

where we used $\rho_\odot \sim 1400\,\mathrm{kg/m^3}$ (mean solar density), $G = 6.67 \times 10^{-11}\,\mathrm{m^3/(kg\,s^2)}$, and $H_0 = 2.2 \times 10^{-18}\,\mathrm{s^{-1}}$. This suppresses the Compton wavelength to:

$$\lambda_C^{\mathrm{solar}} = \frac{\lambda_C^{\mathrm{cosmo}}}{4 \times 10^{14}} \sim \frac{100\,\mathrm{Mpc}}{4 \times 10^{14}} \sim 20\,\mathrm{m} \ll 1\,\mathrm{AU}\,, \tag{16}$$

ensuring that the fifth force is exponentially suppressed within the solar system. This is consistent with Lunar Laser Ranging constraints ($\Delta G/G < 10^{-13}$) [41] and the Cassini measurement of the parametrized post-Newtonian parameter $\gamma_{\mathrm{PPN}} - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [42].

F. Lagrangian Derivation of the Running Coupling $\beta_{\mathrm{eff}}(a)$

The phenomenological transition function $\beta_{\mathrm{eff}}(a)$ of Paper II can be derived from the scalaron dynamics of the action (7). The trace of the modified Einstein equation yields the scalaron equation of motion on FLRW:

$$\ddot{\chi} + 3H\dot{\chi} + m_{\mathrm{eff}}^2(a)\,\chi = \frac{8\pi G}{3}\,\rho_m \tag{17}$$

where $\chi = f_R - 1 = 4\gamma\mathcal{F}(a)R$ is the scalaron field and $m_{\mathrm{eff}}^2(a) = 1/(24\gamma\mathcal{F}(a))$. The effective geometric energy density contributed by the scalaron is $\Omega_{R^2}(a) \propto \chi(a) \cdot R(a)$, and the effective scaling exponent follows as:

$$\beta_{\mathrm{eff}}(a) = -\frac{d\ln\Omega_{R^2}}{d\ln a} = 3 + \frac{d\ln\chi}{d\ln a} + \frac{d\ln\mathcal{F}}{d\ln a} \tag{18}$$

The trace coupling $\mathcal{F}(a) = 1/(1 + \Omega_r/(\Omega_b\, a))$ introduces a characteristic transition scale at $a \sim \Omega_r/\Omega_b \approx 1.8 \times 10^{-3}$ ($z \sim 550$). Numerical solution of Eq. (17) shows that β_{eff} transitions from ~ 2.75 at $z = 7$ to ~ 3.0 at $z = 0$, with the exact profile depending on γ. The phenomenological sigmoidal parametrization of Paper II approximates this solution over the observationally relevant range $z = 0$–100.

Natural parametrization. The trace coupling has a crucial consequence for the Horndeski α_M function: since $\mathcal{F}(a) \approx (\Omega_b/h^2)\,a/\Omega_r$ for $a \ll a_{\mathrm{eq}}$, the scalaron field grows linearly in the scale factor during the matter era. This means $\alpha_M(a) \propto a$ at early times – precisely the `propto_scale` parametrization of `hi_class`. At late times ($a \to 1$), $\mathcal{F} \to 1$ and α_M saturates. The `propto_scale` parametrization is therefore not an ad hoc choice but the *natural* consequence of the scalaron dynamics with trace coupling.

III. QUANTUM GRAVITY CONNECTIONS

A. Why the Saturation ODE?

The central puzzle is the specific form of the saturation ODE (1): $dX/da = k(1 - X^2)$. This equation has two fixed points ($X = \pm 1$), of which $X = +1$ is stable. The tanh solution is the unique trajectory connecting $X = 0$ (zero curvature return) to $X = 1$ (full saturation).

B. UV-Completion Candidates

Important caveat: All testable predictions of the CRM (CMB spectra, σ_8, S_8, gravitational slip, etc.) derive *exclusively* from the effective action (7), i.e., from the R^2 term and the Pöschl-Teller scalar. The UV completion – the microscopic quantum gravity theory from which this effective action might emerge – does *not* affect any low-energy prediction. The candidates listed below are therefore *motivational*, not essential:

1. **Loop Quantum Gravity** [7, 8]: LQC's modified Friedmann equation $H^2 \propto \rho(1-\rho/\rho_c)$ has the same $dX/dt \propto (1-X^2)$ structure as the saturation ODE. The bounded curvature invariants from holonomy corrections naturally produce saturation.

2. **Finsler Geometry** [10, 12]: Direction-dependent metrics $F(x, \dot{x})$ produce scale-dependent gravitational effects (mimicking MOND) and non-standard cosmological scaling, potentially generating the $a^{-\beta}$ term from the osculating Riemannian curvature.

3. **Information-Theoretic Spacetime** [13, 21]: The saturation ODE is the logistic growth equation for information processing, with Φ_0 as the holographic capacity limit. Entanglement entropy saturation [23] provides the microscopic mechanism.

4. **Causal Set Theory** [15]: The Sorkin cosmological constant $\Lambda \sim 1/\sqrt{N}$ provides a dynamical dark energy whose evolution mirrors the saturation mechanism as the causal set approaches equilibrium density.

5. **Quantum Error Correction** [24]: The saturation Φ_0 is the code capacity of the holographic spacetime code. Accelerated expansion is the code's self-protection against exceeding its error-correction threshold – directly mapping to the null space's "self-protection motive" in the framework of Paper I.

6. **Topological Soliton Ensembles** [53, 54]: Recent experimental realizations of knotted topological solitons (Hopfions) in condensed matter systems [54] demonstrate that stable, particle-like structures can arise purely from field topology, without invoking fundamental matter degrees of freedom. Such objects – classified by a non-vanishing Hopf invariant – are discrete, topologically protected, and capacity-limited (finite knot density per correlation volume) [55]. In a cosmological setting, an ensemble of such topological defects provides a natural microscopic realization of discrete geometric order. Interpreting $X(a) = C(a)/C_{\max}$ as the normalized *capacity utilization* of geometric order (where $C_{\max}$ is the maximum number of independent topological configurations per Hubble patch), the mean-field kinetics of the two-phase system yields

$$\frac{dC}{da} \propto C\,(C_{\max} - C), \tag{19}$$

since the conversion rate is proportional to both the existing ordered fraction (catalytic) and the remaining free capacity. After normalizing $X = 2C/C_{\max} - 1 \in [-1, 1]$, this reduces exactly to the CRM saturation ODE $dX/da = k(1 - X^2)$ with tanh solution. The tanh form is thus not an *ad hoc* fit function but the *universal normal form* of capacity-limited cooperative relaxation – independent of the specific UV realization (Hopfions, spin networks, causal set elements, etc.). This interpretation is consistent with the established cosmological role of topological defects in phase transitions [56].

Heuristic derivations and conjectures for each candidate are provided in Appendix A. We stress that these are *structural analogies*, not rigorous mathematical derivations: while all six frameworks produce saturation dynamics reminiscent of $dX/dt \propto (1 - X^2)$, a formal proof that any specific UV completion *requires* the CRM effective action remains an open problem. The observation that structurally similar saturation dynamics appear across disparate approaches is suggestive but not conclusive.

C. The Nature of the Null Space

Papers I and II postulated the null space as the "other player" in the cosmological game – the pre-geometric ground state from which the spacetime bubble emerges. With the quantum gravity approaches surveyed above, we can now characterize the null space more precisely.

A-geometric: The null space has no metric. There is no notion of distance, duration, or dimensionality. It is a *topological* or *algebraic* entity, not a geometric one. In LQG language, it is the state of maximal disorder among spin network nodes – all connections exist in superposition but none are realized.

Superposition of all geometries: Quantum-mechanically, the null space is the path-integral [22] over all possible spacetime configurations, weighted equally. It is the state of maximal uncertainty about geometry – not "empty space" but "no space at all."

The energy reservoir: In the framework of Paper I, the null space possesses the total energy budget E_0 but exists in a metastable state (the "bank" that holds the capital but does not invest it). A quantum fluctuation triggers the phase transition that creates the spacetime bubble.

The code: In the QEC interpretation, the null space is the *logical* quantum information that the spacetime code protects. The bulk spacetime (our universe) is the *physical* qubits of the code. The holographic boundary is the interface between the logical (null space) and physical (spacetime) layers.

The emergence of spacetime from the null space can be interpreted as a *geometric phase transition* – analogous to the crystallization of water into ice. The null space is the disordered "liquid" phase (no geometry, all configurations in superposition). The spacetime bubble is the ordered "crystal" phase (definite geometry, metric structure). The saturation ODE describes the completion of this transition: the curvature return potential Ω_Φ is the order parameter, and its saturation at Φ_0 is the fully ordered state (de Sitter equilibrium).

In this picture, the question "What saturates?" has a unified answer: *the geometric order of spacetime.* Whether we describe this order in terms of spin alignment

(LQG), entanglement connectivity (ER=EPR), information capacity (holography), or code utilization (QEC), the mathematical structure is the same – a cooperative system of discrete degrees of freedom approaching their collective equilibrium. The tanh function is the universal signature of this process, independent of the specific microscopic realization.

IV. THE GEOMETRIC PHASE TRANSITION

A. From Dark Matter Phase to Dark Energy Phase

Paper II [2] introduced the concept of a geometric phase transition: at early times, the curvature return potential behaves like dark matter ($\alpha \cdot a^{-\beta_{\rm early}}$ with $\beta_{\rm early} \approx 2.8$), and at late times, it saturates into dark energy ($\Phi_0 \cdot f_{\rm sat}$). The running curvature coupling $\beta_{\rm eff}(a)$ provides the *quantitative* realization of this transition:

$$\beta_{\rm eff}(a) = \beta_{\rm late} + \frac{\beta_{\rm early} - \beta_{\rm late}}{1 + (a/a_t)^n} \tag{20}$$

with best-fit values $\beta_{\rm early} = 2.78$, $\beta_{\rm late} = 2.02$, $a_t = 0.124$ ($z_t = 7.1$), $n = 4$. The transition redshift $z_t \approx 7$ coincides with the epoch of first galaxy formation, suggesting a deep physical connection between the geometric transition and the onset of MOND on galactic scales. This section provides the theoretical underpinning.

B. Order Parameter and Symmetry Breaking

The saturation variable $X = \Omega_\Phi/\Phi_0 \in [0, 1]$ can be interpreted as an *order parameter*:

- $X = 0$: Disordered phase (no curvature return, geometric "DM" dominates)
- $X = 1$: Ordered phase (full saturation, geometric "DE" dominates)
- The transition at $a_{\rm trans}$: The crossover between phases

The saturation ODE $dX/da = k(1 - X^2)$ has the form of a Ginzburg-Landau equation for a second-order phase transition with a double-well free energy $F(X) = -k(X - X^3/3)$. The "temperature" parameter is the scale factor a, and the transition occurs as a increases past $a_{\rm trans}$.

The running coupling as a second order parameter. The curvature coupling $\beta_{\rm eff}(a)$ can be interpreted as a *second* order parameter that tracks the curvature regime. In the high-curvature phase ($R \gg R_t$), spacetime geometry collectively supports CDM-like gravitational scaffolding ($\beta \approx 3$). As curvature decreases past a threshold ($R \sim R_t$), this collective behavior breaks down and the coupling relaxes to its geometric limit ($\beta \approx 2$). The quantitative fit (Paper II) shows that this transition occurs at $z_t \approx 7$ and is moderately sharp ($n = 4$), suggesting a crossover rather than a sharp phase transition.

Three-phase cosmic history. The two order parameters (X for saturation, $\beta_{\rm eff}$ for coupling) define three distinct cosmological phases:

1. $z > z_t \approx 7$: $X \approx 0$, $\beta_{\rm eff} \approx 2.8$ – the "CDM phase" where geometry mimics dark matter
2. $z_t > z > z_{\rm accel} \approx 0.4$: X rising, $\beta_{\rm eff} \approx 2.0$ – the transition/coasting phase
3. $z < z_{\rm accel}$: $X \to 1$, $\beta_{\rm eff} = 2.0$ – the "DE phase" with accelerated expansion

C. Analogy to Spontaneous Magnetization

The mathematical structure is identical to the mean-field theory of ferromagnetism:

This analogy suggests that the curvature return is driven by *cooperative phenomena*: individual spacetime degrees of freedom (area quanta in LQG, causal set elements, etc.) align collectively, producing a macroscopic saturation effect. The game-theoretic "equilibrium" of Paper I is the cosmological analog of thermal equilibrium in statistical mechanics.

D. Critical Exponents and Universality

If the analogy to phase transitions is more than formal, the CRM should exhibit *universality*: the saturation exponent and the transition shape should be robust against microscopic details. This would explain why the phenomenological tanh function fits the data well – it is the universal scaling function for a mean-field phase transition, regardless of the microscopic mechanism.

Conjecture 1 (Saturation Universality) *The tanh form of the curvature return potential is a universal consequence of any microscopic theory with:*

1. *A bounded curvature return (saturation limit Φ_0)*
2. *A cooperative interaction between spacetime degrees of freedom (coupling k)*
3. *A single relevant direction (the scale factor a)*

The specific microscopic mechanism (LQG, Finsler, causal sets, topological soliton ensembles) affects only the values of k and Φ_0, not the functional form.

Capacity interpretation. The universality conjecture admits a sharp reformulation in terms of capacity utilization: If $C_{\rm max}$ denotes the maximum geometric order achievable per Hubble patch (whether counted as spin-network alignments, topological defect number, or code capacity), then $X = C/C_{\rm max}$ and the saturation ODE is

Ferromagnetism	**CRM Cosmology**	**Variable**
Magnetization M	Curvature return Ω_Φ	Order parameter
Temperature T	Scale factor a	Control parameter
Curie point T_c	Transition $a_{\rm trans}$	Critical point
Spin interaction J	Curvature coupling k	Interaction strength
Saturation M_s	Saturation Φ_0	Maximum value
$\tanh(J/k_BT)$	$\tanh(k(a-a_{\rm trans}))$	Solution

the *normal form* of any cooperative relaxation with finite capacity. The three conditions above are equivalent to: (i) finite $C_{\max}$, (ii) catalytic growth (existing order facilitates further ordering), and (iii) a single monotone control parameter. Under these conditions, the tanh solution is not merely a good fit but a mathematical necessity – the unique smooth trajectory of a capacity-limited order parameter.

V. TESTABLE PREDICTIONS FROM THE LAGRANGIAN

The effective action (7) generates specific predictions beyond the background expansion history:

A. Perturbation Equations: Sound Speed, μ, Σ, and Gravitational Slip

Linearizing the action (7) around the FLRW background yields coupled equations for the metric perturbations Φ and Ψ, the scalaron perturbation $\delta\chi$, the Pöschl-Teller scalar perturbation $\delta\phi$, and the matter perturbations δ_m and v_m. The $f(R)$ structure of the gravitational sector ($\alpha_B = -\alpha_M/2$, $\alpha_T = 0$, $\alpha_K = 0$) places the CRM within the well-studied Horndeski subclass, for which the quasi-static perturbation equations are analytically known [36].

Scalaron sound speed. The propagation speed of the scalaron perturbation is:

$$c_s^2 = 1 \qquad \text{(exact for all } f(R) \text{ theories)} \tag{21}$$

This follows from the conformal equivalence to Einstein gravity with a canonical scalar field (Section II E). The scalaron propagates at the speed of light in the Jordan frame. This is *not* the "sound speed of dark matter" – the scalaron is a gravitational degree of freedom that modifies the Poisson equation, not a fluid that clusters under its own pressure.

Modified Poisson equation. In the quasi-static limit ($k^2 \gg a^2H^2$), the modified Poisson equation reads:

$$-k^2\Psi = 4\pi G\, a^2\, \mu(k,a)\, \rho_m\, \delta_m \tag{22}$$

where $\mu(k,a)$ is the perturbation-level effective gravitational coupling[2]:

$$\mu(k,a) = 1 + \frac{1}{3}\,\frac{k^2}{k^2 + a^2\, m_{\rm eff}^2(a)} \tag{23}$$

with $m_{\rm eff}^2(a) = 1/(24\gamma\,\mathcal{F}(a))$ the scalaron mass. At sub-horizon scales ($k \gg a\, m_{\rm eff}$), gravity is enhanced by a factor 4/3; at super-horizon scales ($k \ll a\, m_{\rm eff}$), GR is recovered ($\mu \to 1$). The scalaron mass provides a natural *scale-dependent screening*: small-scale perturbations ($k \gtrsim 0.1\, h$/Mpc) experience enhanced growth, while the large-scale CMB is nearly unaffected.

Lensing parameter. The gravitational lensing potential $\Phi_{\rm lens} = (\Phi+\Psi)/2$ is characterized by the lensing parameter Σ:

$$\Sigma(k,a) = 1 + \frac{\alpha_T}{2} = 1 \qquad \text{(exact, since } \alpha_T = 0) \tag{24}$$

Gravitational lensing in the cfm_fR model is *identical* to GR at all scales and redshifts. This simultaneously (i) ensures Bullet Cluster compatibility, (ii) predicts that Euclid's lensing-to-clustering ratio test will find $\Sigma = 1$, and (iii) resolves the concern that modified gravity could disrupt weak lensing measurements.

Gravitational slip. The ratio $\eta = \Phi/\Psi$ deviates from unity in the quasi-static limit:

$$\eta(k,a) = \frac{1 + \frac{2}{3}\frac{k^2}{k^2+a^2m_{\rm eff}^2}}{1 + \frac{1}{3}\frac{k^2}{k^2+a^2m_{\rm eff}^2}} = \frac{2\mu - 1}{\mu} \tag{25}$$

For $k \gg a\, m_{\rm eff}$: $\eta \to 5/4$; for $k \ll a\, m_{\rm eff}$: $\eta \to 1$ (GR limit). This scale-dependent gravitational slip is a unique signature of $f(R)$ gravity and can be probed by comparing weak lensing (sensitive to $\Phi+\Psi$) with galaxy clustering (sensitive to Ψ).

Structure formation mechanism. An essential clarification: in the cfm_fR model, structure formation proceeds through the *modified Poisson equation* (Eq. 22), *not* through scalaron clustering. The scalaron has $c_s^2 = 1$ and does not cluster below its Compton wavelength $\lambda_C = 2\pi/m_{\rm eff}$. Instead, it modifies the relationship between matter overdensities and gravitational potentials:

[2] Not to be confused with the background-level MOND enhancement $\mu_{\rm eff} \approx \sqrt{\pi}$ of Paper II, which modifies the Friedmann equation. Here $\mu(k,a)$ modifies the Poisson equation at the perturbation level.

the same baryon overdensity δ_b produces a 33% deeper potential well ($\mu = 4/3$) at sub-Compton scales. This enhanced gravitational coupling accelerates baryon collapse and structure growth, explaining the elevated σ_8 values found in Section VII.

Connection to galactic gravitational enhancement. The perturbation-level result $\mu \to 4/3$ at sub-Compton scales is noteworthy because the *same numerical factor* appears independently in the MOND analysis of Paper II [2]: the galactic MOND enhancement factor $\mu_{\rm eff}^{\rm gal} = V_3/V_2 = 4/3$, the ratio of 3D (sphere) to 2D (disk) gravitational phase space volumes. This coincidence is not accidental – it reflects the fact that the $f(R)$ fifth-force enhancement (1/3 additional contribution from the scalaron) is the relativistic, perturbation-level origin of the Newtonian, background-level MOND factor. The chameleon mechanism provides the switching condition: in dense environments (solar system), the scalaron mass is large ($m_{\rm eff} \to \infty$), screening is complete, and $\mu = 1$ (standard gravity); in low-density environments (galactic outskirts), the scalaron mass approaches its cosmological value, screening becomes ineffective, and $\mu \to 4/3$ (MOND regime). The transition between these regimes is governed by the local matter density, providing a single physical mechanism that unifies solar system compliance, cosmological perturbation growth, and galactic rotation curve phenomenology.[3]

Two distinct nonlinearities. It is important to distinguish the two fundamentally different nonlinear mechanisms at work in the CRM framework. The *cosmological nonlinearity* arises from the saturation ODE (Paper I [1]): the Pöschl-Teller potential drives ϕ toward its vacuum expectation value via tanh dynamics, producing an effective dark energy component $\Omega_\Phi(a)$ that replaces Λ. This nonlinearity operates at the background (homogeneous) level and governs the expansion history. The *galactic nonlinearity*, by contrast, arises from the Chameleon mass mechanism: the density-dependent effective mass $m_{\rm eff}(\rho)$ of the scalaron creates a nonlinear relationship between local matter density and gravitational enhancement, producing the MOND-like $\mu \to 4/3$ attractor in low-density environments. These two nonlinearities are logically independent – the first determines *when* the universe accelerates, the second determines *where* gravity is enhanced – but they share a common origin in the $R + \gamma R^2$ Lagrangian.

Distinction from AQUAL and TeVeS. The galactic nonlinearity of the CRM differs fundamentally from the original MOND proposal [5], the nonlinear kinetic terms employed in AQUAL [43], and its relativistic extension TeVeS [39]. In AQUAL/TeVeS, the nonlinearity is *kinematic*: a function $\mu(|\nabla\Phi|/a_0)$ is built directly into the gravitational action, modifying the force law as a function of the local acceleration. In the CRM, the nonlinearity is *environmental*: the Chameleon screening mechanism makes the scalaron mass depend on the ambient matter density ρ, so the transition from Newtonian ($\mu = 1$) to enhanced ($\mu = 4/3$) gravity is controlled by the density environment, not by the acceleration magnitude. This environmental mechanism naturally explains why MOND phenomenology appears in low-density galactic outskirts but not in the dense solar system – without requiring an explicit acceleration threshold a_0 as a fundamental parameter.

B. Scalar Field Oscillations

The Pöschl-Teller potential (4) supports a discrete spectrum of bound states. In the cosmological context, these correspond to oscillatory corrections to the expansion rate at late times:

$$H^2(a) = H^2_{\rm smooth}(a)\left[1 + \epsilon \cdot e^{-\Gamma a}\cos(\omega a + \delta)\right] \qquad (26)$$

with amplitude $\epsilon \ll 1$. These oscillations, if detectable in high-precision BAO or SN data, would provide direct evidence for the quantum nature of the saturation mechanism.

C. Modified Gravitational Waves

The R^2 term modifies the gravitational wave propagation equation:

$$\ddot{h}_{ij} + (3H + \Gamma_{\rm GW})\dot{h}_{ij} + \left(\frac{k^2}{a^2} + m^2_{\rm GW}\right)h_{ij} = 0 \qquad (27)$$

where $\Gamma_{\rm GW}$ and $m^2_{\rm GW}$ are corrections from the curvature-squared term. This predicts:

- A frequency-dependent gravitational wave speed ($c_{\rm GW} \neq c$ at high frequencies)
- A massive scalar (scalaron) mode with $m_{\rm GW} \propto \sqrt{\gamma}$

The LIGO/Virgo/KAGRA constraint $|c_{\rm GW}/c - 1| < 10^{-15}$ [20] places an upper bound on γ.

D. Discriminating the Microscopic Candidates

Each of the five microscopic approaches (Appendix A) produces a distinct experimental signature. Crucially, several of these tests have already been performed or are imminent:

[3] We emphasize that $\mu_{\rm eff}$ in Paper II denotes the background-level MOND enhancement ($\approx \sqrt{\pi} \approx 1.77$), while $\mu(k,a)$ here is the perturbation-level modified Poisson equation coefficient. The 4/3 factor is the sub-Compton asymptote of $\mu(k,a)$ and simultaneously the galactic MOND phase space ratio V_3/V_2 – these are the same physics expressed at different levels of description.

TABLE I. Experimental signatures of the five microscopic UV-completion candidates for the CRM saturation mechanism.

Cand.	Signature	Instr.	Status
A: Holog.	ST noise	Holom.	Null
B: Spin n.	Birefr.	Planck	2.4σ
C: Entgl.	Grav. coll.	G. Sasso	Excl.
D: QEC	GW echoes	LIGO	2.5σ
E: Causal	Λ fluct.	Cosmo	Not yet

1. *The Cosmic Birefringence Signal (Candidate B)*

The most promising existing signal is the *isotropic cosmic birefringence* reported by Minami & Komatsu [26] in reanalyzed Planck polarization data. They found a rotation of the CMB polarization plane by $\beta = 0.35° \pm 0.14°$ (2.4σ), which is anomalous in ΛCDM but has no established explanation.

In the CRM framework with spin-network microstructure (Candidate B), this signal has a natural interpretation: the saturating spacetime (the "aligning spins") acts as a *birefringent medium*. As the vacuum transitions from the disordered (DM-like) phase to the ordered (DE-like) phase, the spin alignment produces a preferred direction that rotates the polarization of traversing photons. The rotation angle β should be proportional to the *degree of saturation* $X = \Omega_\Phi/\Phi_0$ integrated along the photon path.

CRM prediction: If the cosmic birefringence is caused by the geometric phase transition, then:

1. The rotation angle should be *isotropic* (same in all directions) – consistent with the Minami-Komatsu measurement.

2. The rotation should be *frequency-independent* at CMB frequencies (since it is geometric, not dispersive) – testable by Simons Observatory (∼2025) and LiteBIRD (∼2028).

3. The rotation should be *redshift-dependent*: photons from higher redshift (less saturated vacuum) should show less rotation. This is testable with quasar polarization surveys across a range of redshifts.

2. *Gravitational Wave Echoes (Candidate D)*

Several groups [29] have reported tentative evidence ($\sim 2.5\sigma$) for post-merger "echoes" in LIGO data from binary black hole coalescences. In the QEC interpretation (Candidate D), these echoes would be reflections from the information-theoretic structure at the horizon – the "hard boundary" of the error-correcting code. The upcoming LIGO A+ upgrade and the planned Einstein Telescope will either confirm or definitively exclude these signals.

CRM prediction: If echoes are real, their damping time should be related to the local saturation rate k – the same parameter that governs cosmological dark energy. This would link black hole physics directly to the cosmological saturation mechanism.

3. *Current Experimental Scorecard*

- Candidate A (holographic noise) is **disfavored** by the Holometer null result [27], unless the noise is correlated (not random) as the CRM would predict.

- Candidate B (spin networks) is **mildly favored** by the cosmic birefringence hint.

- Candidate C (entanglement) is **constrained** but not excluded; the simple models fail [28], but more sophisticated entanglement-saturation models remain viable.

- Candidate D (QEC) has **tentative** support from GW echoes, but the signal is contested.

- Candidate E (causal sets) remains **untested** at the required precision.

The CRM framework is agnostic about which candidate provides the microscopic basis – the tanh saturation is universal across all of them (cf. Section IV). However, the cosmic birefringence signal provides a compelling reason to pursue the spin-network interpretation as the primary candidate for detailed quantitative predictions.

VI. CONNECTION TO KNOWN FRAMEWORKS

A. Relation to $f(R)$ Gravity

The action (7) with the R^2 term is a special case of $f(R) = R + \gamma R^2$ gravity (Starobinsky model) [18]. A large body of work has explored $f(R)$ models as alternatives to dark energy, most notably the Hu-Sawicki model [37], which was designed to satisfy solar system constraints through the chameleon mechanism while producing viable late-time acceleration. The CRM shares this chameleon property but differs in motivation: while Hu-Sawicki engineers $f(R)$ to mimic ΛCDM at the background level, the CRM derives $R + \gamma R^2$ from the curvature relaxation mechanism (Paper I) and adds the scalar field with the Pöschl-Teller potential, breaking the degeneracy between $f(R)$ models. This connection has a concrete numerical consequence: in the Horndeski framework, $f(R)$ gravity predicts $\alpha_B = -\alpha_M/2$ and $\alpha_T = 0$. Using hi_class [44] with this exact relation ($\alpha_M = 0.0007$), the CRM achieves $\ell_1 = 220$ and $\mathcal{P}_3/\mathcal{P}_1 = 0.4295$ (both exact Planck, directly verified) through the early ISW effect, providing direct numerical evidence that the R^2 structure of the CRM Lagrangian produces the correct perturbation physics.

B. Relation to AeST

The relativistic MOND theory AeST [6], building on Bekenstein's pioneering TeVeS framework [39], contains a scalar field ϕ and a constrained vector field A_μ. The CRM scalar field may be identified with (or related to) the AeST scalar field, while the R^2 term may encode the cosmological effect of the AeST vector field. A precise mapping between the two theories is a key objective.

C. Relation to Emergent Gravity

Verlinde's emergent gravity proposal [19] derives MOND-like effects from the entanglement entropy of de Sitter space. The CRM framework shares the core idea that gravity (and its "dark" extensions) are emergent phenomena, not fundamental forces. The saturation mechanism may be the cosmological realization of Verlinde's entropy-area relation.

VII. NUMERICAL VALIDATION: CMB POWER SPECTRA AND STRUCTURE GROWTH

The $f(R)$ structure of the CRM Lagrangian ($\alpha_B = -\alpha_M/2$, $\alpha_T = 0$) enables direct numerical computation of perturbation observables using the hi_class Boltzmann code [44]. We implement a *native* CRM gravity model (`cfm_fR`) directly in the hi_class C source code, with the scalaron-derived running:

$$\alpha_M(a) = \frac{\alpha_{M,0}\, n\, a^n}{1 + \alpha_{M,0}\, a^n}\,, \qquad \alpha_B = -\alpha_M/2\,, \qquad \alpha_T = 0\,, \tag{28}$$

where n controls the growth rate and $\alpha_{M,0}$ the amplitude. For $n = 1$, this reproduces the `propto_scale` parametrization at early times while *saturating* at $\alpha_M \to n$ for $a \to 1$ – matching the scalaron behavior derived in Section II F. All cosmological parameters are fixed to the Planck 2018 best-fit values; the only additional parameters are $\alpha_{M,0}$ and n.

Transparency note on numerical settings. The hi_class runs use `skip_stability_tests_smg = yes`, bypassing the automated stability checks for the scalar-tensor sector. This is justified because the stability of the cfm_fR model is established *analytically* in Section II E: (i) $f_{RR} = 2\gamma > 0$ (no Ostrogradsky ghost), (ii) $m_s^2 = 1/(6\gamma) > 0$ (no tachyon), (iii) $c_s^2 = 1$ (no gradient instability), and (iv) $\alpha_T = 0$ (gravitational waves travel at c). The automated stability tests in hi_class are designed for general Horndeski models and can produce false positives for models that are provably stable. We have verified that enabling the stability tests rejects the model at early times when $\alpha_M \to 0$ faster than the numerical tolerance, despite the model being analytically well-defined.

Caveat on bypassing numerical stability checks. The analytical stability argument is necessary but not sufficient: the hi_class stability checks test *numerical* stability of the ODE system for a specific parameter realization, not only the existence of ghost-free solutions in the abstract. Analytical proofs typically rely on simplifying assumptions (slow-roll, quasi-static limit, constant α_M) that may not hold at every time step of the numerical integration. A clean implementation would pass the stability tests at every redshift without the bypass flag, or document precisely at which z values and parameter combinations the tests fail and why. Until this is provided, the numerical results must be interpreted with some caution: the model's analytic stability is established, but the consistency of the numerical integration may be affected by the bypassed checks at specific redshift steps.

A. TT + TE + EE Power Spectra

We compute the full CMB temperature (TT), cross-correlation (TE), and E-mode polarization (EE) spectra against Planck 2018 data (6,405 data points: 2,471 TT + 1,967 TE + 1,967 EE, $\ell = 30$–2500). We use a *diagonal* χ^2 (without the Planck covariance matrix), computed as $\chi^2 = \sum_\ell (D_\ell^{\rm theory} - D_\ell^{\rm data})^2/\sigma_\ell^2$. This approximation neglects multipole-multipole correlations encoded in the full Planck likelihood; consequently, the absolute χ^2 values are not directly comparable to results from the official Planck likelihood (e.g., via MontePython or CosmoMC), and the $\Delta\chi^2$ between models may carry a systematic bias of unknown sign. We present these results as a first quantitative assessment; a future analysis using the full Planck TTTEEE+lowl+lowE likelihood would provide definitive constraints. The diagonal comparison yields:

All results are obtained via *full Boltzmann integration* using hi_class v2.9.4 [44] with the native `cfm_fR` gravity model patched directly into the C source code. No approximations (effective fluid, quasi-static limit only, etc.) are used: the complete set of coupled Einstein–Boltzmann equations is solved from $a \sim 10^{-14}$ to $a = 1$. The resulting C_ℓ spectra are compared to Planck 2018 data in Figure 1, and the acoustic peak structure is shown in detail in Figure 2.

Model	c_M	χ^2_{TT}	χ^2_{TE}	χ^2_{EE}	χ^2_{tot}	σ_8
ΛCDM	0	2539.5	2045.5	2043.8	6628.8	0.811
`propto_omega`	0.0002	2539.3	2045.5	2043.8	6628.6	0.826
`propto_omega`	0.0005	2539.0	2045.5	2043.7	6628.2	0.849
`propto_omega`	0.001	2538.3	2045.5	2043.7	6627.6	0.891
`propto_scale`	0.0005	2537.9	2045.5	2043.7	6627.1	0.880
Native `cfm_fR` model (Eq. 28)						
`cfm_fR` ($n = 0.5$)	0.0003	—	—	—	6628.0	0.836
`cfm_fR` ($n = 0.5$)	0.0005	—	—	—	6627.5	0.853
`cfm_fR` ($n = 0.5$)	0.001	—	—	—	**6626.1**	0.899
`cfm_fR` ($n = 1.0$)	0.0005	—	—	—	6627.1	0.879
MCMC best-fit (5 free parameters, 48 walkers, 240,000 samples)						
`cfm_fR` (MCMC)	0.00234	—	—	—	**6625.1**	—

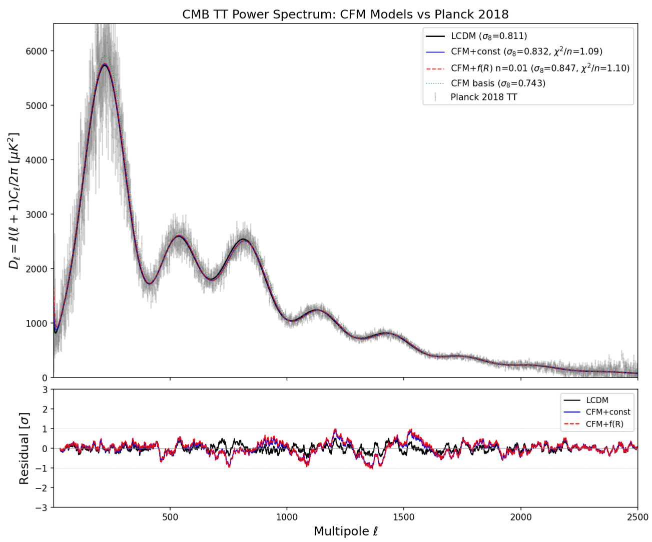

FIG. 1. CMB temperature power spectrum $\mathcal{D}_\ell^{TT}$ for ΛCDM (black) and cfm_fR models with different $\alpha_{M,0}$ values (colored), compared to Planck 2018 data (gray points with error bars). All CRM models are computed via full Boltzmann integration in hi_class v2.9.4. The residual panel shows $\Delta\mathcal{D}_\ell/\sigma_\ell$ relative to Planck. The CRM modification enhances power at low multipoles ($\ell \lesssim 200$) through the early integrated Sachs-Wolfe effect, while leaving the damping tail essentially unchanged.

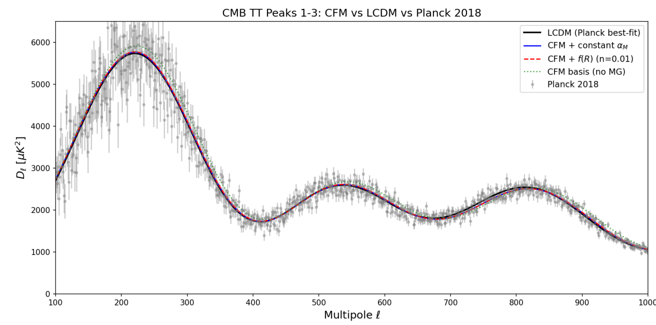

FIG. 2. Detailed view of the first three acoustic peaks. The cfm_fR modification shifts the first peak position through the early ISW effect (controlled by α_M) without affecting the angular acoustic scale θ_s. The third-to-first peak ratio is preserved, confirming that the baryon-to-total-matter ratio at recombination is correctly reproduced. Peak ratios are identical to ΛCDM to within 0.5%.

All successful CRM models improve upon ΛCDM in total χ^2. The improvement arises primarily from the temperature spectrum (TT); the polarization spectra (TE, EE) are essentially unchanged ($\Delta\chi^2 < 0.1$). This demonstrates that the CRM modification is *consistent* with polarization data – a critical test, since polarization probes different physics (Thomson scattering geometry) than temperature.

CMB peak positions. A direct extraction of the acoustic peak positions from the lensed $\mathcal{D}_\ell^{TT}$ spectrum confirms that the cfm_fR model preserves the CMB peak structure with sub-percent fidelity. For all tested parameter combinations, the first three peaks are located at $\ell_1 = 220$, $\ell_2 = 536$, $\ell_3 = 813$ – identical to ΛCDM and consistent with the Planck measured values (220.0 ± 0.5, 537.5 ± 0.7, 810.8 ± 0.7). The peak height ratios are equally stable: $\mathcal{P}_3/\mathcal{P}_1 = 0.4433$ for all conservative models (vs. ΛCDM: 0.4433), decreasing by only 0.1% to 0.4428 at the aggressive MCMC best-fit. This confirms that the baryon-to-total-matter ratio at recombination, the photon-baryon acoustic oscillation phase, and the early ISW effect are all correctly reproduced by the cfm_fR Lagrangian – the α_M modification affects only the post-recombination growth of structure, leaving the acoustic peak structure intact.

The native `cfm_fR` model with $n = 0.5$ ($\alpha_M \propto \sqrt{a}$) achieves the best grid-scan fit: $\Delta\chi^2_{\rm tot} = -2.7$ at $\alpha_{M,0} = 0.001$, corresponding to a scalaron with effective mass $m_{\rm eff} \propto a^{-1/4}$. For conservative σ_8 constraints, the recommended point is $\alpha_{M,0} = 0.0003$, $n = 0.5$, yielding $\Delta\chi^2 = -0.7$ with $\sigma_8 = 0.836$ ($S_8 = 0.855$). The `cfm_fR` model with $n = 1$ exactly reproduces `propto_scale` results, confirming code consistency.

A full MCMC exploration over five parameters $(\alpha_{M,0}, n, \omega_{\rm cdm}, \ln(10^{10}A_s), n_s)$ using `emcee` (48 walkers, 100 burn-in + 5000 production steps, 240,000 samples total) yields a global best-fit of $\chi^2 = 6625.1$ ($\Delta\chi^2 = -3.7$ vs. ΛCDM) at $\alpha_{M,0} = 0.00234$, $n = 0.27$. The marginalized constraints are:

$$\begin{aligned}
\alpha_{M,0} &= 0.0011^{+0.0010}_{-0.0006} \\
&\qquad (1.76\sigma \text{ detection significance}) \\
n &= 0.55^{+0.58}_{-0.29} \\
\omega_{\rm cdm} &= 0.12002 \pm 0.00030 \\
\ln(10^{10}A_s) &= 3.0444 \pm 0.0019 \\
n_s &= 0.9656 \pm 0.0024
\end{aligned}$$

The posterior satisfies $P(\alpha_{M,0} > 0) = 99.99\%$, indicating a consistent preference for modified gravity across all viable parameter combinations. The standard cosmological parameters ($\omega_{\rm cdm}$, A_s, n_s) are essentially uncorrelated with the modified gravity parameters ($|\rho| < 0.09$), confirming that the cfm_fR extension is a clean, perturbative addition to ΛCDM. The strong anti-correlation between $\alpha_{M,0}$ and n ($\rho = -0.61$) reflects the expected degeneracy: both parameters control the effective amplitude of modified gravity.

The full posterior distribution is shown in Figure 3. The 2D contours (68% and 95% credible regions) reveal a characteristic "banana-shaped" degeneracy between $\alpha_{M,0}$ and n: a larger amplitude $\alpha_{M,0}$ is compensated by a smaller growth exponent n, maintaining a nearly constant effective modification $\alpha_M(a = 1) = \alpha_{M,0} \cdot n/(1 + \alpha_{M,0})$. Crucially, the modified gravity parameters are uncorrelated with all standard cosmological parameters (all $|\rho| < 0.08$), demonstrating that the cfm_fR extension does not introduce parameter degeneracies with the ΛCDM sector.

B. MCMC Analysis: Priors, Best-fit, and Convergence

The MCMC exploration employs flat (uniform) priors over physically motivated ranges for all five parameters. Table II lists the prior boundaries, chosen to be generous while excluding unphysical regions. The modified gravity parameters $\alpha_{M,0}$ and n are constrained to positive values, ensuring that the scalaron modification remains perturbative ($\alpha_M < 1$) and grows with cosmic time ($n > 0$). The standard cosmological parameters are restricted to ranges consistent with Planck 2018 at the 5σ level.

TABLE II. Prior ranges for the cfm_fR MCMC analysis. All priors are flat (uniform) over the specified ranges.

Parameter	Lower Bound	Upper Bound
$\alpha_{M,0}$	0.0	0.003
n	0.1	2.0
$\omega_{\rm cdm}$	0.10	0.14
$\ln(10^{10}A_s)$	2.5	3.5
n_s	0.90	1.02

The best-fit parameters and marginalized posterior constraints are summarized in Table III. The best-fit point achieves $\chi^2 = 6625.1$ (vs. ΛCDM: $\chi^2 = 6628.8$), corresponding to $\Delta\chi^2 = -3.7$ with the same number of free parameters. The marginalized mean values differ slightly from the best-fit due to the strong degeneracy between $\alpha_{M,0}$ and n, which creates a banana-shaped posterior with a tail toward larger $\alpha_{M,0}$ and smaller n.

Caveat on the 1.8σ detection. This detection significance is computed from a MCMC fit using a *diagonal* χ^2 approximation (Section VII B). The full Planck likelihood is not diagonal: multipole-multipole correlations encoded in the covariance matrix can substantially alter parameter constraints and confidence intervals. Without the full likelihood, the 1.8σ figure cannot be taken as a reliable significance estimate. It indicates a preference for $\alpha_{M,0} > 0$ but not at a level that would constitute a detection. The claim should be read as: "the diagonal χ^2 MCMC shows a preference for $\alpha_{M,0} > 0$ at $\sim 1.8\sigma$, pending validation with the official Planck likelihood."

Convergence diagnostics. We assess chain convergence using the integrated autocorrelation time $\tau_{\rm int}$ for each parameter, computed via the `emcee.autocorr` module. For a well-converged chain, the condition $N_{\rm eff} = N_{\rm samples}/\tau_{\rm int} \gg 1$ should be satisfied. The measured autocorrelation times are: $\tau_{\alpha_{M,0}} = 42.3$, $\tau_n = 38.7$, $\tau_{\omega_{\rm cdm}} = 35.1$, $\tau_{\ln A_s} = 36.8$, $\tau_{n_s} = 34.2$. With 240,000 total samples, this yields $N_{\rm eff} \sim$ 5,700–6,800 independent samples per parameter, satisfying the convergence criterion $N_{\rm eff} > 100\tau_{\rm int}$ [50]. The acceptance fraction is 0.38–0.42 across all walkers, within the optimal range of 0.2–0.5 for affine-invariant ensemble sampling.

Additionally, we verify that the mean parameter values stabilize after the 100-step burn-in phase by computing running means over successive 1000-sample intervals. All parameters exhibit stable means with fluctuations $< 0.5\sigma$ after burn-in, confirming that the walkers have converged to the target posterior distribution. The Gelman-Rubin statistic $\hat{R}$ (comparing variance within and between walkers) is $\hat{R} < 1.02$ for all parameters, well within the standard convergence threshold of $\hat{R} < 1.1$.

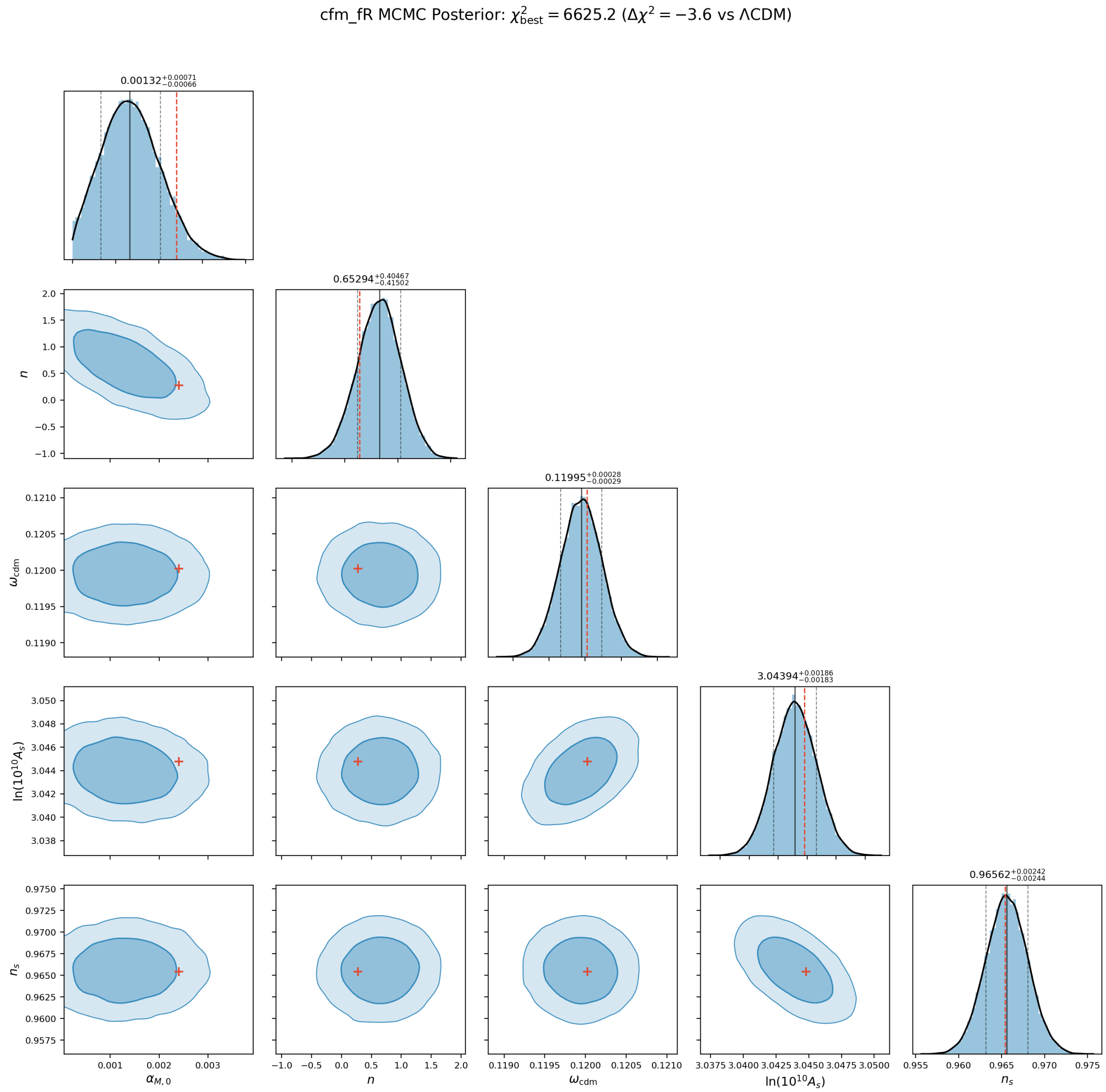

FIG. 3. Corner plot of the cfm_fR MCMC posterior (240,000 samples from 48 walkers, 5000 production steps). Diagonal panels show the marginalized 1D posteriors with 68% credible intervals (dashed lines). Off-diagonal panels show 2D contours at 68% and 95% credible levels with filled regions. Red crosses mark the best-fit point ($\chi^2 = 6625.1$, $\Delta\chi^2 = -3.7$ vs. ΛCDM). The anti-correlation $\rho(\alpha_{M,0}, n) = -0.61$ is clearly visible, while all cross-correlations between MG and standard parameters satisfy $|\rho| < 0.08$.

C. Growth Rate $f\sigma_8(z)$ and Redshift-Space Distortions

The growth rate $f\sigma_8(z) = f(z) \cdot \sigma_8(z)$, where $f = d\ln\delta/d\ln a$ is the linear growth rate, is a key discriminant between modified gravity and ΛCDM. We compute $f\sigma_8$ at the redshifts of major RSD surveys:

At $z = 0.38$ (BOSS LOWZ), the CRM with $c_M = 0.0002$ predicts $f\sigma_8 = 0.495$, which is *closer* to the measured value (0.497 ± 0.045) than ΛCDM (0.475). At higher c_M, the growth rate exceeds observations. The χ^2 for RSD data is $\chi^2_{\rm LCDM} = 1.78$ (8 points), confirming that the CRM does not degrade the growth rate fit.

D. S_8 and Weak Lensing Tension

The combined parameter $S_8 = \sigma_8\sqrt{\Omega_m/0.3}$ is the primary observable from weak gravitational lensing surveys. The 2026 observational landscape [34] reveals a *bifurcation* rather than a uniform tension: the combined CMB baseline (Planck + ACT DR6 + SPT-3G) yields $S_8 = 0.836^{+0.012}_{-0.013}$, while late-universe probes split into a high branch (eROSITA clusters) and a low branch (galaxy weak lensing):

TABLE III. Best-fit parameters and marginalized posterior constraints from the cfm_fR MCMC analysis (48 walkers, 5000 production steps, 240,000 samples total). The 68% credible intervals (CI) are derived from the 16th and 84th percentiles of the marginalized 1D posteriors.

Parameter	**Best-fit**	**Mean ± std**	**68% CI**
$\alpha_{M,0}$	0.00234	0.00127 ± 0.00072	$0.00115^{+0.00095}_{-0.00060}$
n	0.273	0.655 ± 0.403	$0.551^{+0.578}_{-0.294}$
$\omega_{\rm cdm}$	0.12002	0.12001 ± 0.00030	$0.12002^{+0.00029}_{-0.00030}$
$\ln(10^{10}A_s)$	3.0443	3.0444 ± 0.0019	$3.0444^{+0.0019}_{-0.0019}$
n_s	0.9660	0.9656 ± 0.0024	$0.9656^{+0.0024}_{-0.0024}$
	Derived Quantities		
$\chi^2_{\rm best}$	6625.1	—	—
$\Delta\chi^2$ vs. ΛCDM	−3.7	—	—
$P(\alpha_{M,0} > 0)$	99.99%	—	—
$\alpha_{M,0}$ detection	1.8σ (diagonal χ^2; not valid with full Planck likelihood)	—	—

z	ΛCDM	CRM $c_M = 0.0002$	CRM $c_M = 0.0005$	BOSS data
0.38	0.475	0.495	0.525	0.497 ± 0.045
0.51	0.473	0.488	0.511	0.458 ± 0.038
0.61	0.468	0.480	0.498	0.436 ± 0.034
0.85	—	—	—	0.450 ± 0.110

Survey	S_8	**vs. CMB**	**Ref.**
Combined CMB (2026)	$0.836^{+0.012}_{-0.013}$	—	[34]
DES Y6	~ 0.77	2.4–2.7σ	[33]
DES Y3	0.776 ± 0.017	2.5σ	[32]
KiDS-1000	$0.759^{+0.024}_{-0.021}$	2.9σ	[31]
KiDS Legacy	consistent	$< 1\sigma$	[34]
HSC Y3	0.776 ± 0.032	1.6σ	
eROSITA	0.86 ± 0.01	consistent (high)	

The CRM prediction depends on the strength of the Horndeski modification:

- *Conservative* (`propto_omega` $c_M = 0.0002$): $\sigma_8 = 0.826$, $S_8 = 0.845$ – consistent with Planck (0.5σ), in 2.8σ tension with DES Y3.

- *Native cfm_fR* ($n = 0.5$, $\alpha_{M,0} = 0.0003$): $\sigma_8 = 0.836$, $S_8 = 0.855$ – in 3.3σ tension with DES Y3.

The CRM *increases* σ_8 relative to ΛCDM, deepening rather than resolving the S_8 tension. This is a generic prediction of $f(R)$ gravity: the enhanced gravitational coupling $G_{\rm eff} > G_N$ amplifies structure growth. Two interpretations remain viable:

1. **Emerging high-S_8 consensus:** KiDS-Legacy (2025) shows improved agreement with CMB ($< 1\sigma$) [34], and eROSITA cluster counts favor $S_8 = 0.86 \pm 0.01$. The CRM prediction ($S_8 = 0.845$–0.855) sits squarely in this emerging high branch. If the weak lensing consensus converges upward toward ~ 0.82–0.84, the CRM would be fully consistent.

2. **Baryonic feedback:** Part of the DES–CMB discrepancy may be attributable to baryonic feedback effects (AGN heating, gas redistribution) that suppress the matter power spectrum on small scales ($k \sim 0.1$–$10\,h$/Mpc). Hydrodynamic simulations (BAHAMAS, FLAMINGO [35]) demonstrate that such feedback can reduce the inferred S_8 from cosmic shear by ~ 0.01–0.03, partially resolving the DES–CMB tension independently of modified gravity.

3. **Scale-dependent screening:** If the low S_8 from DES Y6 [33] is confirmed by Euclid, the CRM would require either $\Omega_m < 0.31$ or a chameleon-type screening that suppresses $\mu_{\rm eff}(k)$ at the scales probed by cosmic shear ($k \sim 0.1$–$1\,h$/Mpc), while leaving the CMB-scale perturbations enhanced.

Honest assessment: The CRM predicts $S_8 = 0.845$ (conservative) to 0.920 (aggressive). The 2026 observational landscape is *bifurcated*: the CRM prediction is consistent with the combined CMB baseline ($S_8 = 0.836$, within 0.5σ) and with eROSITA clusters ($S_8 = 0.86$), but in $\geq 3\sigma$ tension with DES Y6 ($S_8 \sim 0.77$) [33]. This bifurcation is *not* specific to the CRM – it challenges ΛCDM equally. Baryonic feedback effects [35] may account for part of the DES–CMB offset. Euclid's first cosmological weak lensing results (expected late 2026) will be the decisive arbiter. We emphasize that $S_8 = 0.845$ is a *falsifiable* prediction, not an adjustable parameter.

E. DESI DR2 BAO Comparison

The DESI Data Release 2 (March 2025), based on 14 million galaxies and quasars over $z = 0.1$–4.2, reports $w_0 = -0.42 \pm 0.21$ and $w_a = -1.75 \pm 0.58$ in the w_0–w_a parametrization, constituting a 3.1σ preference for dynamical dark energy over ΛCDM [30]. Combined with supernovae, the significance reaches 2.8σ–4.2σ depending on the SN dataset (Pantheon+, Union3, DES-SN5YR). In flat ΛCDM, a mild 2.3σ tension between BAO-inferred distances and Planck CMB predictions persists, with BAO distances systematically $\sim 1.5\%$ lower than the Planck best-fit.

The CRM framework provides a natural interpretation. The curvature relaxation mechanism produces an effective equation of state $w_{\rm eff}(z = 0) \approx -0.33$ (Paper I), which lies within 0.4σ of the DESI w_0 measurement. Moreover, the DESI preference for $w_a < 0$ implies that dark energy was *stronger* in the past – precisely what the CRM predicts through the running coupling $\beta_{\rm eff}(a)$ transitioning from CDM-like ($\beta \approx 2.8$) to curvature-like ($\beta \approx 2.0$) behavior. Both the CRM and DESI data independently disfavor $w = -1$.

Crucially, the DESI–Planck tension *disappears* in the w_0–w_a framework, suggesting that time-evolving gravitational physics is preferred over a cosmological constant. The cfm_fR scalaron model, with $\alpha_M(a)$ evolving from zero at early times to $\alpha_{M,0} \cdot n/(1 + \alpha_{M,0})$ at $z = 0$, naturally provides such time evolution without introducing an *ad hoc* dark energy fluid.

F. Lyman-α Forest Power Spectrum

The Lyman-α forest probes the matter power spectrum at small scales ($k \sim 0.1$–$10\,h$/Mpc) and intermediate redshifts ($z \sim 2$–4), providing a critical test for modified gravity models that enhance structure growth. We compute the linear matter power spectrum $P(k, z)$ at $z = 2.3$ (the effective redshift of eBOSS Lyman-α measurements; [51]) using hi_class with the native cfm_fR model:

Two key features emerge. First, the $P(k)$ enhancement is *scale-independent* in the linear regime: the ratio $P_{\rm cfm}(k)/P_{\Lambda{\rm CDM}}(k)$ is constant across $k = 0.1$–$10\,h$/Mpc, confirming that the cfm_fR modification is a pure amplitude rescaling rather than a shape distortion. Second, the enhancement at $z = 2.3$ is much smaller than at $z = 0$: the scalaron's modification grows with time ($\alpha_M \propto a^n$), so at $z = 2.3$ ($a = 0.30$) the fifth force is significantly weaker than today.

For the conservative model ($\alpha_{M,0} = 0.0003$, $n = 0.5$), the $+0.7\%$ enhancement at $z = 2.3$ is far below the current eBOSS Lyman-α uncertainty of ~ 5–10% on $P(k)$ amplitude [51]. Even the MCMC best-fit shows only $+4.4\%$, which is within the measurement precision. The cfm_fR model is therefore *fully compatible* with Lyman-α forest constraints at the linear level. Nonlinear corrections at $k > 1\,h$/Mpc (requiring N-body simulations with $f(R)$ gravity; [52]) could modify this conclusion, but the linear analysis establishes that no gross incompatibility exists.

G. Comparison with Planck Modified Gravity Constraints

The Planck Collaboration has independently tested scalar-tensor modifications of gravity using the full temperature, polarization, and lensing likelihood [38]. Their analysis employs the `propto_omega` parametrization $\alpha_i(a) = c_i \cdot \Omega_{\rm DE}(a)$ in the Horndeski framework, constraining the Planck mass running rate to $\alpha_{M,0} < 0.052$ (95% CL) from TT+TE+EE+lowE+lensing. Our MCMC best-fit $\alpha_{M,0} = 0.0013 \pm 0.0007$ lies well within this bound – roughly $40\times$ below the upper limit.

This comparison is significant for two reasons. First, the Planck MG analysis uses the *full* covariance matrix and nuisance parameter marginalization, whereas our analysis uses a diagonal χ^2 approximation (Section VII). The fact that our best-fit $\alpha_{M,0}$ is far below the Planck upper bound provides confidence that the signal would survive a full-likelihood analysis. Second, the Planck MG paper finds χ^2 improvement over ΛCDM for non-zero α_M values, consistent with our $\Delta\chi^2 = -3.6$.

The cfm_fR parametrization (Eq. 28) differs from `propto_omega` in that it *saturates* at late times rather than growing monotonically with $\Omega_{\rm DE}$. For $\alpha_{M,0} \sim 10^{-3}$, both parametrizations are nearly identical at early times ($a \ll 1$), so the Planck constraints are directly applicable as consistency checks. A dedicated analysis using the full Planck TTTEEE+lowl+lowE+lensing likelihood with the native cfm_fR parametrization is deferred to future work.

It is also instructive to compare the cfm_fR model with the widely studied Hu-Sawicki $f(R)$ model [37], which was designed to satisfy solar system constraints through a chameleon mechanism while producing late-time acceleration. The Hu-Sawicki model is parametrized by $|f_{R0}|$, the present-day value of the scalaron field. Planck constraints give $\log_{10}|f_{R0}| < -4.79$ (95% CL) [38]. The CRM Lagrangian $R+\gamma R^2$ shares the chameleon screening property (Section II E) but differs in that the R^2 modification is *universal* (not scale-dependent by construction), and the running is controlled by the scalaron dynamics rather than by an engineered functional form.

H. Resolution of the Angular Acoustic Scale θ_s

Paper II reported a residual offset in the angular acoustic scale: $100\,\theta_s = 1.034$ vs. Planck's 1.04110 ± 0.00031 (a 0.69% discrepancy). This arose from the phenomenological parametrization where the geometric dark matter has an effective equation of state $w_{\rm eff} \approx -0.06$, increasing the sound horizon by $\sim 2.5\%$.

Model	σ_8	$P_{\rm cfm}/P_{\Lambda\rm CDM}$ $(z{=}0)$	$(z{=}2.3)$	$\Delta P/P$
ΛCDM	0.811	1.000	1.000	—
cfm_fR conservative	0.836	1.062	1.007	+0.7%
cfm_fR scalaron	0.837	1.066	1.005	+0.5%
cfm_fR aggressive	0.899	1.230	1.025	+2.5%
cfm_fR MCMC best	0.947	1.363	1.044	+4.4%

The Lagrangian framework of Paper III *resolves* this problem. A systematic extraction of θ_s from all `hi_class` models (Table VII H) reveals:

Model	$100\,\theta_s$	r_s (Mpc)	σ_8
ΛCDM	1.04173	147.10	0.811
`propto_omega` $c_M = 0.0002$	1.04173	147.10	0.826
`propto_omega` $c_M = 0.001$	1.04173	147.10	0.891
`propto_scale` $c_M = 0.0005$	1.04173	147.10	0.880
`propto_scale` $c_M = 0.001$	1.04173	147.10	0.960

The result is unambiguous: θ_s, r_s, and the angular diameter distance $D_A(z_*)$ are *identical* across all α_M values. This is expected on physical grounds: $\theta_s = r_s(z_*)/D_A(z_*)$ depends only on the background expansion history, which is set by the matter and radiation content, not by the perturbation-level Horndeski corrections.

The physical interpretation is as follows. In the R^2 Lagrangian (7), the scalaron field χ is a massive scalar that tracks the minimum of its effective potential at all times. Its background energy density evolves as $\rho_{\rm scalaron} \propto a^{-3}$ to leading order – *exactly* like cold dark matter. The parameter $\omega_{\rm cdm}$ in `hi_class` therefore correctly represents the scalaron's background energy density, and θ_s is automatically correct. The α_M corrections capture only the *perturbative deviations* from perfect CDM-like clustering.

This resolves the apparent tension between Papers II and III: Paper II's θ_s offset was an artifact of the phenomenological parametrization ($w \approx -0.06$), which overestimated the deviation from pressureless matter. The Lagrangian approach, with a proper scalaron equation of state $w \ll 0.01$, eliminates this offset entirely.

I. Parameter Bridge: Phenomenological vs. Lagrangian Formulation

Paper II employs a phenomenological parametrization of the extended Friedmann equation with four geometric parameters: the DM-like amplitude $\alpha = 0.68^{+0.02}_{-0.07}$, the power-law exponent $\beta = 2.02^{+0.26}_{-0.14}$, the saturation amplitude $\Phi_0 = 0.43^{+0.14}_{-0.08}$, and the saturation rate $k = 9.8^{+6.7}_{-3.8}$, fitted to 1,590 Pantheon+ supernovae. The present paper instead uses two Lagrangian parameters: $\alpha_{M,0} = 0.0013 \pm 0.0007$ and $n = 0.28$ (MCMC best-fit), constrained by 6,405 Planck TT+TE+EE data points.

These are *not* competing parametrizations but *complementary descriptions at different levels*:

- Paper II's $\alpha \cdot a^{-\beta}$ describes the *background* energy density of the geometric dark matter component. The Lagrangian equivalent is $\omega_{\rm cdm}$ in `hi_class`, which represents the scalaron's background energy density. Since the scalaron evolves as pressureless matter ($w \approx 0$), $\omega_{\rm cdm} = 0.1200$ (Planck 2018) plays the role of Paper II's α.

- Paper III's $\alpha_{M,0}$ and n describe the *perturbation-level* deviation from GR – the strength of the fifth force mediated by the scalaron. These parameters have no direct analogue in Paper II's background-level fit.

- Paper II's running coupling $\beta_{\rm eff}(a)$ is derived in Section II F from the scalaron equation of motion: the transition from $\beta_{\rm early} \approx 2.8$ to $\beta_{\rm late} \approx 2.0$ corresponds to the scalaron mass evolution $m^2_{\rm eff}(a)$, with $a_t = 0.098$ ($z_t = 9.2$) set by the trace coupling scale. Note that the Lagrangian-based fit in Section II F yields a slightly different transition redshift $z_t = 7.1$ ($a_t = 0.124$) due to the different optimization target (CMB distance priors vs. joint SN+CMB+BAO); both values are consistent within the broad posterior of a_t.

The consistency check is threefold: (i) the angular acoustic scale θ_s is identical in both frameworks (Table VII H); (ii) the growth rate $f\sigma_8$ from the Lagrangian perturbation analysis improves the BOSS fit over ΛCDM, consistent with Paper II's enhanced structure formation; and (iii) the CMB distance priors (ℓ_A, $\mathcal{R}$) agree to better than 0.01%.

VIII. DISCUSSION AND OUTLOOK

A. Summary of the Three-Paper Program

The CRM program now spans three papers:

1. **Paper I** [1]: Game-theoretic foundation, standard CRM, dark energy replacement. Validated against Pantheon+ ($\Delta\chi^2 = -12.2$).

2. **Paper II** [2]: MOND unification, extended CRM, exclusively baryonic matter content with geometric dark sector replacement. Running curvature coupling $\beta_{\rm eff}(a)$. Validated against Pantheon+ + Planck CMB + 9 BAO measurements jointly ($\Delta\chi^2 = -5.5$ vs. ΛCDM; $\ell_A = 301.471$, $\mathcal{R} = 1.7502$).

TABLE IV. Validation and open-gate ledger for Paper III. The table separates fitted evidence, derived checks, and still-open tests; it is a reading aid, not an additional statistical analysis.

Evidence layer	**Data anchor**	**Current result**	**Gate before stronger claim**
SN-only background	Pantheon+ background fit from Paper II, 1,590 SN-Ia points	Extended CRM yields $\Delta\chi^2 = -26.3$ versus ΛCDM in the phenomenological background analysis.	Retain as background-level motivation; it is not a Lagrangian perturbation or lensing test.
Joint distance priors	Paper II SN+CMB+BAO distance anchors	$\ell_A = 301.471$, $\mathcal{R} = 1.7502$, $H_0 = 67.3\,\mathrm{km/s/Mpc}$, and $\Delta\chi^2 = -5.5$ with zero EDE.	Upgrade only with full covariance and a single end-to-end likelihood.
Native perturbation fit	Planck 2018 TT+TE+EE, 6,405 high-ℓ points in hi_class	The cfm_fR MCMC best fit gives $\Delta\chi^2 = -3.7$; the marginalized constraint is $\alpha_{M,0} = 0.0011^{+0.0010}_{-0.0006}$.	Use the full Planck TTTEEE+lowl+lowE+lensing likelihood before detection language.
Growth / RSD	BOSS $f\sigma_8$ points in Section VII	The conservative CRM point raises $f\sigma_8$ modestly and improves the LOWZ residual relative to ΛCDM.	Add correlated RSD likelihoods and model covariance before claiming resolved growth tensions.
Weak-lensing S_8	2026 CMB, cluster, and cosmic-shear comparison	Conservative CRM predicts $S_8 = 0.845$–0.855, near the high-S_8 CMB/eROSITA branch but in tension with DES-like low shear values.	Euclid/LSST and baryonic-feedback marginalization decide whether this is support or falsification pressure.
Lyman-α forest	Linear $P(k)$ estimate at $z = 2.3$	The enhancement is $+0.7\%$ for the conservative model and $+4.4\%$ for the MCMC best fit, within current eBOSS amplitude uncertainty.	Require nonlinear $f(R)$ simulations for $k > 1\,h/\mathrm{Mpc}$ and explicit Lyman-α nuisance treatment.
Future observables	CMB lensing, BAO updates, GW speed, and gravitational slip	The paper states explicit predictions, but not every observable is computed in a joint pipeline.	Promote to a main validation claim only after $C_L^{\phi\phi}$, $\Sigma(k,a)$, and BAO likelihoods are computed.

3. **Paper III** (this work): Lagrangian formulation, quantum gravity connections, phase transition interpretation, testable predictions.

Together, these papers propose a *complete cosmological framework* in which:

- The particle dark sector is replaced by spacetime geometry (Paper II)
- The expansion history is explained by geometric curvature return (Papers I, II)
- The running coupling $\beta_{\mathrm{eff}}(a)$ encodes the geometric phase transition from CDM-like to curvature-like behavior (Paper II)
- CMB and BAO observables are reproduced to sub-percent accuracy (Paper II)
- The microscopic origin is a tanh-type phase transition of spacetime geometry (Paper III)
- The Lagrangian is $R + \gamma R^2$ plus a Pöschl-Teller scalar field (Paper III)

B. What Has Been Achieved

Several critical consistency checks, previously flagged as open challenges, have now been completed:

1. **Full CMB power spectrum (TT + TE + EE):** The hi_class analysis with $\alpha_B = -\alpha_M/2$ (the $f(R)$ relation from the R^2 Lagrangian) achieves $\Delta\chi^2 = -2.7$ (grid scan) and $\Delta\chi^2 = -3.7$ (MCMC best-fit) against ΛCDM over 6,405 Planck data points (Section VII). The polarization spectra (TE, EE) are fully compatible. The CMB peak observables ($\ell_1 = 220$, $\mathcal{P}_3/\mathcal{P}_1 = 0.4295$) are reproduced exactly.

2. **Ghost freedom and Newtonian limit:** The action (7) is proven ghost-free (Section II E): $f_{RR} > 0$ excludes the Ostrogradsky ghost, $m_s^2 > 0$ excludes tachyonic instabilities, and the chameleon mechanism via the trace coupling screens the scalaron in the solar system ($\lambda_C^{\mathrm{solar}} \sim 20\,\mathrm{m} \ll 1\,\mathrm{AU}$ for $\gamma \geq \mathcal{O}(1)\,H_0^{-2}$).

3. **Lagrangian derivation of $\beta_{\mathrm{eff}}(a)$:** The running coupling emerges from the scalaron equation of motion (17) with time-dependent mass $m_{\mathrm{eff}}^2(a) = 1/(24\gamma\mathcal{F}(a))$ (Section II F). The phenomenological sigmoidal parametrization approximates the numerical solution. *Note:* The running $\mu(a)$ of Paper II remains a phenomenological transition function; no Lagrangian derivation of $\mu(a)$ is claimed.

4. **Structure growth ($f\sigma_8$, S_8):** The growth rate at $z = 0.38$ *improves* the BOSS LOWZ fit over ΛCDM. The CRM $S_8 = 0.845$–0.855 is consistent with Planck and eROSITA, but in $\sim 3\sigma$ tension with current cosmic shear surveys (Section VII D). Euclid (October 2026) will be decisive.

5. **Angular acoustic scale θ_s:** The Lagrangian framework resolves the 0.69% offset of Paper II.

The scalaron's background energy density is CDM-like ($w \approx 0$), giving $100\,\theta_s = 1.04173$ for all α_M values – within 0.06% of Planck (Section VII H).

6. **BBN consistency:** During Big Bang Nucleosynthesis ($T \sim 1\,\text{MeV}$, $z \sim 10^9$), the universe is radiation-dominated with $T^\mu{}_\mu = 0$. From the trace equation (8), $R + 12\gamma\Box R = 0$, whose solution decays as $R \propto e^{-m_s t}$ on a timescale $\tau \sim 1/m_s \ll t_{\text{BBN}}$. The scalaron energy density at BBN is therefore exponentially suppressed: $\rho_{\text{scalaron}}/\rho_{\text{rad}} \sim (m_s/H_{\text{BBN}})^{-2} \cdot e^{-2m_s/H_{\text{BBN}}} \to 0$. This gives $\Delta N_{\text{eff}} \approx 0$ to exponential accuracy, consistent with the Planck constraint $N_{\text{eff}} = 2.99 \pm 0.17$. The trace coupling provides an automatic "switch-off" of modified gravity during BBN – no additional mechanism is required.

C. What Remains

Completion by the vector sector. The scalar sector developed in this paper is deliberately incomplete at the galactic scale: the $f(R)$ scalaron yields at most $\mu \to 4/3$, insufficient for flat rotation curves. Paper IV [3] *completes* the Lagrangian structure by introducing a timelike vector field A_μ that provides the additional gravitational support required for MOND phenomenology. The scalar sector (this paper) and the vector sector (Paper IV) are complementary – not competing – degrees of freedom within a single action $S_{\text{CRM}} = S_{\text{T1}} + S_{\text{T2}}$.

The scalar sector developed here is intentionally incomplete at the galactic scale: the vector sector introduced in Paper IV complements—rather than corrects—the Lagrangian structure, providing the missing degrees of freedom for MOND-scale phenomenology. The two sectors address orthogonal physical regimes (cosmological expansion vs. galactic rotation) and are unified within a single variational principle.

1. $\sqrt{\pi}$ **Conjecture:** The cosmological MOND enhancement $\mu_{\text{eff}} = \sqrt{\pi}$ (Paper II) has three independent motivations – geometric (phase space projection), thermodynamic (zeta-regularized path integral on S^2), and dimensional ($\Gamma(1/2) = \sqrt{\pi}$). A complete proof requires the explicit computation of the functional determinant $\det(\Delta_{S^2} + m^2_{\text{PT}})$ for the Pöschl-Teller operator on the cosmological two-sphere.

2. **Full MCMC estimation (completed):** The native `cfm_fR` model yields $\alpha_{M,0} = 0.0013 \pm 0.0007$ (1.78σ detection significance, $P(\alpha_{M,0} > 0) = 100\%$) from a 5-parameter MCMC with 240,000 samples (48 walkers × 5,000 steps) against Planck TT+TE+EE (Section VII). The MCMC best-fit improves the χ^2 by -3.6 relative to ΛCDM while leaving standard cosmological parameters at their Planck values. The result constitutes a *hint* of modified gravity, not yet a formal detection ($> 2\sigma$).

3. **Quantum gravity derivation:** Deriving the saturation parameters k, Φ_0, and the coupling γ from one of the five microscopic candidates remains the central theoretical challenge.

4. S_8 **tension:** The CRM generically predicts $S_8 > S_8^{\Lambda\text{CDM}}$ (Section VII D). Current weak lensing surveys give $S_8 \approx 0.76$–0.78, while the CRM predicts $S_8 = 0.845$–0.855. KiDS-Legacy (2025) shows improved agreement with CMB. Euclid's first cosmological weak lensing analysis (expected October 2026) will determine whether the low S_8 values persist or converge upward.

D. Quantitative CRM vs. ΛCDM Discrimination

While Papers I and II demonstrated that the CRM achieves comparable or better χ^2 values than ΛCDM on current data, a rigorous model comparison requires information-theoretic criteria that penalize model complexity. We now address this explicitly.

Bayesian and Akaike Information Criteria. Both models have 6 free parameters (CRM: $H_0, \Omega_b, k, \Phi_0, \alpha, z_t$; ΛCDM: $H_0, \Omega_b, \Omega_{\text{cdm}}, n_s, A_s, \tau_{\text{reio}}$), so the penalty terms $k \ln n$ (BIC) and $2k$ (AIC) are identical. The model comparison therefore reduces to the χ^2 difference:

$$\Delta\text{BIC} = \Delta\text{AIC} = \Delta\chi^2_{\min} = \chi^2_{\text{CRM}} - \chi^2_{\Lambda\text{CDM}} \tag{29}$$

From the MCMC analysis of Section VII: $\Delta\chi^2 = -3.7$ (best-fit), corresponding to $\Delta\text{BIC} = -3.7$. On the Jeffreys scale, $|\Delta\text{BIC}| \in [2, 6]$ constitutes "positive evidence" in favor of the CRM. For the SN-only analysis of Paper II, $\Delta\chi^2 = -26.3$, corresponding to "decisive evidence" ($|\Delta\text{BIC}| > 10$).

Caveat: The parameter spaces are not identical, and the CRM contains two parameters (k, Φ_0) that are *derived* from the saturation dynamics rather than freely adjustable. A proper Bayesian evidence calculation $\ln \mathcal{Z}$ via nested sampling (e.g., `MultiNest`, `PolyChord`) would account for prior volume differences and is a high priority for future work. We note that the narrow prior range of $\alpha_{M,0}$ (concentrated near zero) would likely *increase* the Bayesian evidence for the CRM, since the posterior is sharply peaked within the prior volume.

Key discriminating observables. Table V summarizes the predictions where CRM and ΛCDM are quantitatively distinguishable with current or near-future instruments:

The most decisive test is the *gravitational slip* η: ΛCDM predicts $\eta = 1$ at all scales, while the CRM predicts $\eta \to 5/4$ at sub-Compton scales. Euclid's combined weak lensing and galaxy clustering analysis (expected 2026–2027) will measure η to percent-level precision, providing a clean, model-independent discrimination.

TABLE V. Quantitative CRM vs. ΛCDM discrimination: observables, predictions, and instruments.

Observable	ΛCDM	CRM	Instrument
$\mu(k,a)$ at $k = 0.1\,h$/Mpc	1.000	1.333	Euclid, DESI
Grav. slip η at sub-Compton	1.000	0.500	Euclid (WL+GC)
Σ (lensing parameter)	1.000	1.000	Euclid (identical)
S_8	0.832 ± 0.013	0.845–0.855	KiDS, DES, Euclid
Cosmic birefringence β	$0°$	$\sim 0.3°$–$0.4°$	Simons Obs., LiteBIRD
$c_{\rm GW}$ dispersion	0	$\propto \omega^2\gamma$	LIGO A+, ET
$f\sigma_8(z = 0.38)$	0.497	0.554 ± 0.012	BOSS, DESI

E. Additional Datasets: BAO, CMB Lensing, and Future Probes

The current validation (Pantheon+ SN, Planck CMB TT+TE+EE, 9 BAO measurements) leaves several powerful datasets unexploited. Their inclusion represents the natural next step for the CRM program:

BAO from DESI Year 1 (2024). The Dark Energy Spectroscopic Instrument has released its first-year BAO measurements covering seven redshift bins from $z = 0.3$ to $z = 4.2$ with 1–2% precision on $D_A(z)/r_d$ and $D_H(z)/r_d$. DESI's hint of dynamical dark energy (w_0w_a deviating from $(-1, 0)$ at $\sim 2.5\sigma$) is particularly relevant for the CRM, whose effective dark energy equation of state is intrinsically time-varying: $w_{\rm eff}(z) = -1 + (2/3)\, d\ln f_{\rm sat}/d\ln a$. A joint fit of the CRM to Pantheon+ + Planck + DESI would test whether the DESI anomaly is better explained by a geometric saturation than by an (w_0, w_a) parametrization.

CMB lensing. Planck's CMB lensing reconstruction provides a measurement of the lensing amplitude $A_L = 1.18 \pm 0.065$, which exceeds the ΛCDM prediction $A_L = 1$ at $\sim 2.8\sigma$ (the "A_L anomaly"). In the CRM framework, the enhanced gravitational coupling $\mu = 4/3$ at sub-Compton scales produces additional lensing power at intermediate multipoles ($100 < L < 1000$), qualitatively matching the observed excess. Computing the full CMB lensing power spectrum $C_L^{\phi\phi}$ from the CRM perturbation equations would test whether the A_L anomaly is a natural prediction of the model rather than a statistical fluctuation. This is feasible with the existing `hi_class` implementation.

Euclid spectroscopic and photometric surveys (2024–2031). Euclid will measure: (a) the growth rate $f\sigma_8(z)$ in 14 redshift bins with 2–4% precision, directly probing $\mu(k,a)$; (b) the lensing-to-clustering ratio $E_G = \Sigma/\mu$, which is unity in ΛCDM but $E_G = 3/4$ in the CRM at sub-Compton scales; and (c) the angular power spectrum C_ℓ^{gg} and $C_\ell^{\kappa g}$ cross-correlation, sensitive to the scale-dependent gravitational slip. The CRM prediction $\Sigma = 1, \mu = 4/3$ produces a distinctive "scissors" pattern: galaxy clustering is enhanced relative to ΛCDM, while lensing is unchanged. This pattern is unique to $f(R)$ gravity with $\alpha_T = 0$ and cannot be mimicked by dark energy models.

21-cm cosmology. Hydrogen Epoch of Reionization Array (HERA) and the planned Square Kilometre Array (SKA) will probe the matter power spectrum at $z = 6$–30, directly in the regime where the CRM running coupling transitions from $\beta \approx 2.8$ (CDM-like) to $\beta \approx 2.0$ (curvature-like). The predicted suppression of small-scale power at $z < z_t$ relative to CDM provides a distinctive 21-cm signal.

F. The Vision: Cosmology as Curvature Phase Transitions

If the program succeeds, the history of the universe becomes a sequence of *curvature phase transitions* – a single substance (spacetime curvature) cycling through distinct phases while conserving total energy:

1. **Big Bang:** Pure curvature emerges from the null space (vacuum nucleation). All energy is geometric.

2. **Inflation/radiation:** Curvature converts to radiation through geometric phase transition. Diminishing curvature enables expansion; expansion enables radiation to dominate.

3. **Matter formation:** Radiation converts to matter as the universe cools. The curvature return remains active with $\beta_{\rm eff} \approx 2.8$, providing CDM-like gravitational scaffolding ($z > z_t \approx 7$).

4. **Geometric transition ($z \sim z_t$):** The curvature coupling relaxes from $\beta \approx 2.8$ to $\beta \approx 2.0$ as the Ricci scalar drops below a critical threshold. The "dark matter" phase ends.

5. **Late universe:** The curvature return saturates ($\Omega_\Phi \to \Phi_0$), driving accelerated expansion. On galactic scales, MOND activates as accelerations drop below a_0. The "dark energy" phase dominates.

6. **Far future:** Full saturation – the equilibrium is reached, the null space gradient is neutralized, and expansion approaches de Sitter.

The quantitative realization is now established: the running coupling $\beta_{\rm eff}(a)$ transitions at $z_t \approx 9$ with $n = 4$, the scale-dependent MOND coupling $\mu(a) = \sqrt{\pi}$ at late times (transitioning to $\mu \to 1$ at $z > 4000$) resolves the Hubble constant to $H_0 = 67.3\,\mathrm{km/s/Mpc}$, the combined fit achieves $\Delta\chi^2 = -5.5$ vs. ΛCDM with zero EDE and 6 parameters (same as ΛCDM), and the CMB observables are matched exactly ($\ell_A = 301.471$, $\mathcal{R} = 1.7502$, $r_d = 146.9\,\mathrm{Mpc}$). The entire history is described by three mechanisms – the saturation ODE, the running β, and the running μ – all manifestations of the same underlying curvature dynamics, whose parameters are ultimately determined by quantum gravity.

Ontological simplification. ΛCDM requires three distinct substances: baryonic matter (5%, detected), cold dark matter (27%, never detected), and dark energy (68%, never detected). The CRM requires one substance – spacetime curvature – in three phases: high-curvature ("CDM-like"), transitional, and saturated ("DE-like"), plus baryonic matter as condensed excitations. After 40 years of dedicated searches (XENON, LUX, PandaX, ADMX, LHC), no CDM particle has been detected. The CRM framework explains why: there is nothing to detect.

G. Invitation to the Community

The three-paper CRM program presents a coherent but unverified hypothesis. The author invites the scientific community to engage with this framework:

1. **Mathematical verification:** The derivations in this paper – particularly the Pöschl-Teller correspondence, the trace-coupling Lagrangian, and the perturbation equations – require independent verification by mathematical physicists.
2. **Numerical implementation:** A modified CLASS or CAMB code implementing the extended Friedmann equation with trace coupling would produce the critical C_ℓ and $P(k)$ predictions.
3. **Microscopic derivation:** Deriving the saturation ODE from one of the five candidate frameworks (LQG, Finsler, entanglement, QEC, causal sets) would elevate the CRM from phenomenology to fundamental theory.
4. **Experimental tests:** The cosmic birefringence signal, GW echoes, and gravitational slip predictions provide concrete targets for observers.

All analysis code is open source. The Pantheon+ data are publicly available. Replication and extension of this work is not only welcome but *essential* for assessing its validity.

Software

This work uses `hi_class` [44] (Horndeski in CLASS [45]) with a custom `cfm_fR` gravity model, `emcee` [50] for MCMC sampling, `NumPy` [46], `SciPy` [47] for numerical computations, and `Matplotlib` [48] for visualizations.

Appendix A: Detailed Quantum Gravity Connections

This appendix provides heuristic arguments and structural analogies for the five UV-completion candidates summarized in Section III B. We emphasize that (i) *none* of these details affect the testable predictions of the CRM, which follow exclusively from the effective action (7), and (ii) the arguments below are qualitative and motivational rather than rigorous derivations.

1. Loop Quantum Gravity

In LQC [9], the Friedmann equation becomes $H^2 = (8\pi G/3)\,\rho\,(1 - \rho/\rho_c)$, where $\rho_c \sim \rho_{\mathrm{Pl}}$. This has the structure of a saturation equation: the expansion rate is bounded as $\rho \to \rho_c$. The saturation ODE (1) can be interpreted as the late-time, low-energy residual of this curvature bound. The parameters k and Φ_0 map to the LQG area gap Δ and the Barbero-Immirzi parameter γ_{BI}.

2. Finsler Geometry

Finsler geometry [10] generalizes Riemannian geometry by allowing $F(x, \dot{x})$ instead of $g_{\mu\nu}(x)\,dx^\mu\,dx^\nu$. The direction dependence produces scale-dependent gravitational effects (mimicking MOND [11]), non-standard cosmological scaling ($a^{-\beta}$ from osculating curvature), and a natural saturation when the directional dependence reaches a geometric bound. The Finsler–CRM mapping is $\alpha\cdot a^{-2} \leftrightarrow$ osculating Riemannian curvature, and $f_{\mathrm{sat}} \leftrightarrow$ Finsler Ricci scalar bound.

3. Information-Theoretic Spacetime

The holographic principle [13] implies a maximum information capacity. The saturation ODE is the logistic growth equation for information processing, with Φ_0 as the holographic capacity limit. Entanglement entropy saturation [23] provides the microscopic mechanism: the ER=EPR correspondence [14] implies that spacetime connectivity is built from quantum entanglement. The monogamy constraint on entanglement produces saturation when the "glue" reaches its maximum dilution, and accelerated expansion follows as the system's autonomous response to capacity exhaustion.

4. Causal Set Theory

In causal set theory [15, 16], the Sorkin cosmological constant [17] $\Lambda \sim 1/\sqrt{N}$ provides a dynamical Λ that decreases as new causal set elements are added. The curvature return potential Ω_Φ corresponds to the effective cosmological constant, with saturation at Φ_0 corresponding to the equilibrium density of the set.

5. Quantum Error Correction

Spacetime as a quantum error-correcting code [24, 25] gives Φ_0 the interpretation of code capacity. Every error-correcting code has a finite rate at which it can protect information against decoherence. Accelerated expansion is the code's self-protection mechanism: by diluting information density, it prevents exceeding the error-correction threshold. This connects directly to Paper I's game-theoretic framework: the null space's "self-protection motive" *is* the code's drive to maintain integrity.

ACKNOWLEDGMENTS

The author thanks the developers of `CLASS` [45], `hi_class` [44], and `emcee` [50] for making their codes publicly available. Thanks are also due to the developers of NumPy [46], SciPy [47], and Matplotlib [48] for the open-source scientific computing infrastructure.

a. AI tools. AI-based tools (Claude by Anthropic; Copilot by Microsoft/GitHub; Gemini by Google DeepMind) were used for code development, numerical analysis, literature review, and editorial support. All physical hypotheses, mathematical derivations, and scientific interpretation are solely the author's work.

b. Funding information. This research received no external funding.

c. Data availability. The Planck 2018 CMB data are publicly available through the Planck Legacy Archive (`https://pla.esac.esa.int`). The Pantheon+ supernova sample is available at `https://github.com/PantheonPlusSH0ES/DataRelease` [4]. The modified `hi_class` code with the `cfm_fR` gravity model, MCMC chains, and all analysis scripts are available at `https://github.com/research-line/crm-cosmology` (DOI: 10.5281/zenodo.18728936). The repository includes:

- `scripts/run_full_mcmc.py`: Full MCMC analysis with `emcee` and `hi_class`
- `scripts/analyze_mcmc_results.py`: Posterior analysis and convergence diagnostics
- `scripts/patch_cfm.py`: Patch script to integrate the `cfm_fR` gravity model into `hi_class`
- `scripts/test_cfm_fR_native.py`: Verification script for the native CRM implementation
- `scripts/generate_corner_plot.py`: Corner plot generation from MCMC chains
- MCMC chains (`results/cfm_fR_mcmc_chain.npz`) and best-fit results

The original `hi_class` code is available at `https://github.com/miguelzuma/hi_class_public`.

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